LANGUAGE LEARNING
A Journal of Research in Language Studies

Language Learning ISSN 0023-8333

L2 Vocabulary Learning From Reading: Explicit and Tacit Lexical Knowledge and the Role of Learner and Item Variables

Irina Elgort and Paul Warren

Victoria University of Wellington

This study investigates acquisition of second language (L2) vocabulary from reading a connected authentic text. Advanced and upper-intermediate L2 (English) participants read a long expository text for general understanding, with embedded critical vocabulary items (pseudowords). Explicit knowledge of the critical items was examined using a meaning generation task, while their tacit knowledge was probed using form and semantic priming in lexical decision tasks. Results revealed a complex landscape of contextual L2 word learning in which individual differences (age, L2 lexical proficiency, first language, gender, learning strategies, levels of enjoyment) and lexical and text characteristics (concreteness, frequency, distribution, and saliency of use) individually and together affect L2 lexical development from reading. Implications of these results for contextual L2 word learning are discussed.

Keywords vocabulary; reading; tacit knowledge; explicit knowledge; priming; strategies; age; proficiency

Introduction

Reading is a key source of second and foreign language (L2) input for adult language learners. Readers at a range of L2 proficiencies commonly come across unknown words while reading, but it is not clear to what degree they take advantage of these incidental encounters in building their L2 lexicons. The questions of how new L2 words are learned incidentally and what factors affect

---

We thank the anonymous reviewers for their comments on previous versions of this manuscript. We also extend special thanks to reviewer Scott Crossley, whose comments have helped us refine our data analysis. Thanks also to Anna Piasecki for her assistance with data collection and to Victoria University of Wellington for the University Research Funding grant awarded to this project.

Correspondence concerning this article should be addressed to Irina Elgort, Victoria University of Wellington, P.O. Box 600, Wellington 6012, New Zealand. E-mail: Irina.Elgort@vuw.ac.nz

DOI: 10.1111/lang.12052

this learning are key in building our understanding of L2 lexical development from reading. Incidental word learning from reading is a complex process that is affected by learner, text, and word variables (Pulido, 2007). Controlled experimental studies have greatly advanced knowledge about how these different variables affect contextual word learning (e.g., Ferrel Tekmen & Daloğlu, 2006; Pulido, 2007; Pulido & Hambrick, 2008; Vidal, 2011; Webb, 2008). Studies that explore word learning in long connected L2 texts under naturalistic conditions, on the other hand, are more scarce (Horst, Cobb, & Meara, 1998; Pellicer-Sanchez & Schmitt, 2010), and the few that exist have often been limited by challenges of design and measurement. Particularly challenging is the question of how vocabulary knowledge gain ought to be documented in contextual, less controlled designs. Our position is that both explicit and tacit measures of word knowledge should be used in order to get a more comprehensive understanding of the quality of lexical knowledge established for contextually learned words.

The present study is an exploration into early stages of contextual L2 (English) word learning that sought to address these issues. Building on experimental research into variables that affect contextual word learning, this study observes the interplay of these variables in the naturalistic conditions of meaning-focused reading of a long authentic text. Methodologically novel aspects of this study are its attention to both explicit word knowledge (via a meaning generation task) and tacit word knowledge (via form and semantic priming) and the use of mixed-effects modeling to explore the effects of participant and item variables on learning.

Factors Affecting Incidental Word Learning From Reading

Incidental word learning from reading is a complex process that is affected by lexical, text/context, learner, and task factors (e.g., Pulido, 2007; Fukkink & de Glopper, 1998). General theories of learning suggest that frequency of encounters is one of the most powerful predictors of learning (Anderson, 1982, 1989; Ellis, 2002; Hulstijn, 2001). This frequency-based view of learning also holds for contextual word learning: the number of times a new word occurs in the text (i.e., rate of occurrence) is a key factor affecting this type of learning (Brown, Waring, & Donkaewbua, 2008; Ferrel Tekmen & Daloğlu, 2006; Horst et al., 1998; Pellicer-Sanchez & Schmitt, 2010; Pigada & Schmitt, 2006; Saragi, Nation, & Meister, 1978; Waring & Takaki, 2003; Webb, 2007; Zahar, Cobb, & Spada, 2001). Vidal (2011), for example, reported that rate of occurrence accounted for 47% of variation in vocabulary gain. Little detectable learning has been observed in L2 vocabulary studies for the initial occurrences of a new word in context, with some gains reported for 6–10 occurrences and a further increase in word knowledge above 10 exposures (Pellicer-Sanchez & Schmitt, 2010). These findings show that incidental word learning from reading is an incremental process that happens over time. Word learning in the first language (L1) and L2 is also affected by such lexical factors as word concreteness, familiarity, and frequency (Ellis & Beaton, 1993; see Crossley, Salsbury, McNamara, & Jarvis, 2011, for an overview).

The quality of the context in which new words occur also affects learning. When L2 vocabulary is learned from sentence contexts, the presence of more contextual clues that make it easier to guess the meanings of unknown words results in significantly higher scores in tests of recognition and recall of meaning (Webb, 2008). Availability of contextual cues to meaning is also beneficial in L1 word learning, with semantically constraining contexts (i.e., contexts in which the word is easy to predict) being more conducive to learning than low-constraint contexts (Borovsky, Kutas, & Elman, 2013; Frishkoff, Collins-Thompson, Perfetti, & Callan, 2008; Frishkoff, Perfetti, & Collins-Thompson, 2010; Mestres-Misse, Rodriguez-Fornells, & Munte, 2007). On the other hand, some L2 contextual word learning studies suggest that words that are easy to guess are also easy to forget (e.g., Mondria & Wit-de Boer, 1991; Hulstijn & Laufer, 2001). One explanation of these conflicting findings is that they arise as a result of the nature of the task performed by the reader, as predicted by the task-induced involvement hypothesis proposed by Hulstijn and Laufer (2001). Reading a continuous text for general understanding, for example, creates conditions under which the reader may be less likely to engage in contextual word learning, as long as the meaning of the text overall is understood (Pulido, 2007). Therefore, one needs to distinguish between inferring word meanings for text understanding and for learning (Lawson & Hogben, 1996); contextual word learning is more likely to occur if the reader notices vocabulary items as new forms and attempts to link these forms with their inferred meaning. This brings us to the effects of individual learner differences on word learning from reading.

Even though participants in incidental word learning studies are traditionally instructed to read for meaning, individual readers may approach unfamiliar words differently (i.e., some may try to actively guess their meaning from context, while others may not; e.g., Fraser, 1999; Nassaji, 2003; Paribakht & Wesche, 1999). L2 readers, in particular, are likely to have developed individual strategies for dealing with new L2 words (Nation, 2001), which may affect learning.

The lexical proficiency of individual readers also affects contextual word learning in L1 and L2. Horst et al. (1998) reported a weak positive correlation between vocabulary level scores (representing participants' L2 lexical proficiency) and incidental vocabulary gain. Ferrel Tekmen and Daloğlu (2006) found that readers in their high-vocabulary-level group learned significantly more words than those in the two lower level groups. Pulido and Hambrick (2008) found that L2 readers with larger passage sight vocabularies (i.e., an ability to efficiently decode and recognize words in a reading passage) were better at vocabulary learning through comprehension. In L1 studies, similar positive effects of vocabulary knowledge and reading ability have been observed. For example, Frishkoff et al. (2008) found that deliberate vocabulary learning from context was more successful for those who scored higher on the Nelson-Denny Vocabulary and Nelson-Denny Reading Comprehension tests (Brown, Nelson, & Denny, 1973).

Finally, text comprehension is another factor in contextual word learning. Moderate to strong positive correlations between text recall (as a measure of comprehension) and L2 vocabulary gain and retention were reported by Rott (1999), when new words were embedded in brief narrative texts. Pulido (2007) also found positive correlations between text comprehension (i.e., percentage of the total number of semantic propositions correctly recalled by participants) and incidental vocabulary development from reading, measured in terms of intake (recognition memory of forms) and gain/retention (recognition/production of meaning). In Pulido's study, nonsense L2 words were embedded in contrived script-based narrative passages and participants were asked to recall each story in their L1 after reading it. The correlation between text comprehension and recognition of lexical form in Pulido's study was qualified by topic familiarity, with the effect of comprehension diminishing for familiar topics. This suggests that, when they are reading for general understanding, readers are less likely to pay attention to unfamiliar words if the overall meaning of the passage is well understood.

Incidental L2 Word Learning From Long Connected Texts

In studies on word learning from context, it is common to use short contrived passages or single sentence contexts (e.g., Pulido, 2007, 2009; Webb, 2005, 2007, 2008). This approach makes it easier to control and manipulate text/context factors affecting learning. It is not clear, however, whether L2 word learning observed in short simplified passages or stand-alone sentences is the same as incidental vocabulary acquisition from reading long connected texts. Laufer and Shmueli (1997), for example, observed that shorter contexts were more effective for L2 word learning than longer contexts, suggesting that the text length is itself a factor in learning. In a study using a graded reader, Zahar et al. (2001) found no effect of contextual richness (measured using the Beck, McKeown, & McCaslin, 1983, model of contextual support) on vocabulary gain, over and above that of rate of occurrence. This suggests that naturalistic word learning is a complex process driven not only by individual factors, but also by interrelations between them. With this in mind, we argue that controlled experimental studies that investigate how specific learner, text, and lexical factors affect contextual word learning need to be supplemented with studies that explore word learning in long connected L2 texts, outside the laboratory.

To our knowledge, there are few L2 studies that deal with vocabulary acquisition from long authentic texts, and most of them have limitations related to their design and measures of vocabulary knowledge gain. In an earlier incidental vocabulary acquisition study (Saragi et al., 1978) participants read an authentic L1 text (60,000 words) that included novel words from an invented language, *nadsad*. However, reading in their L1 made the whole experience more like learning unknown L1 words. In a study conducted by Horst et al. (1998), L2 learners followed along while the teacher read a simplified version of one of Thomas Hardy's novels (21,000 words) in class, over 10 days. However, this reading-along experience is different from independent silent reading. Another study (Pellicer-Sanchez & Schmitt, 2010) used an authentic L2 (English) text (67,000 words) and assessed recognition and recall of spelling, meaning, and grammatical class of novel words. However, these novel words were from an African language unknown to the L2 readers (i.e., Idemili dialect of the Nigerian language Ibo), rather than from the L2. Thus, critical words in this study (e.g., *Ekwenzu*, *Nneka*, *Ogbanje*, *Iyi-uwa*) did not meet orthographic, phonological, or morphological patterns of the L2, presumably making the learning of their form more difficult. In addition, the researchers avoided using critical words in nontransparent contexts, which reduced the authenticity of the learning experience. In the next section we address another important limitation of contextual L2 word learning studies, that is, their approaches to measuring gain.

Measures of Learning

In studies of word learning, a key decision is how to measure learning outcomes. Waring and Takaki (2003) demonstrate that reported word learning outcomes vary depending on the type of test used. Learners achieve better scores in tests of receptive than productive knowledge, in recognition than in recall tests, and in prompted (e.g., multiple-choice) than in unprompted tests (e.g., cloze tests, free production; Brown et al., 2008; Pellicer-Sanchez & Schmitt, 2010; Pigada & Schmitt, 2006; Webb, 2007). The learning may also proceed differently for different knowledge components, such as knowledge of form and knowledge of meaning. Pellicer-Sanchez and Schmitt (2010), for example, found that recognition of meaning developed faster than recognition of form (although learning of form may have been more difficult in this study because critical words were Ibo rather than English, the participants' L2).

Traditionally, L2 studies of vocabulary learning use tests that require some form of controlled retrieval of a word's form or meaning, such as multiple-choice, cloze, and translation tests. In completing these tasks, learners are able to use explicit task strategies and resort to resource-intensive cognitive processes (e.g., using mnemonics and other higher-order cognitive strategies) to retrieve meaning from form or vice versa. Conversely, fluent language use, such as reading, depends critically on fast and fluent processing of lower-level information (i.e., fast and accurate processing of a word's form and retrieval of its meaning). This fluent efficient lexical processing frees up readers' cognitive resources to be allocated to higher-level processing needs, in order to make sense of connected text. L1 studies (Andrews, 2008; Andrews & Bond, 2009; Andrews & Hersch, 2010; Castles, Davis, Cavalot, & Forster, 2007) show that good readers have higher-quality (more fully specified) orthographic, phonological, and semantic representations of known words than poor readers and are able to retrieve these representations reliably, consistently, and synchronously (Perfetti & Hart, 2001, 2002; Perfetti, 2007). Similarly, inadequate low-level processing can be detrimental to comprehension in L2 (Grabe, 2009). Representational knowledge is tacit, that is, language users are not able to report or control it (e.g., language users cannot explain why it is harder to retrieve a known word when it is presented after another word that is very similar to it in its form). This representational knowledge, underpinning fluent L2 lexical processing, can be assessed using word-processing tasks (usually performed under time pressure) that utilize priming to foreground the constituent lexical representations. We argue, therefore, that both explicit and tacit measures of word knowledge should be used to get a full picture of the quality of lexical knowledge established for recently learned words.

The Present Study

The present study set out to examine: (1) lexical knowledge that develops incidentally for novel L2 words through reading a long connected text, including both explicit and tacit knowledge, and (2) how this knowledge is affected by participant and item variables. Following Hulstijn's (2013) definition of incidental learning, we use the term to describe "the acquisition of a word or expression without the conscious intention to commit the element to memory, such as 'picking up' an unknown word from listening to someone or from reading a text" (p. 2632). Going beyond traditional approaches adopted by studies on L2 vocabulary learning from reading, both explicit and tacit knowledge of critical items is measured in this study, using a battery of tasks.

To evaluate participants' explicit knowledge of critical vocabulary items (pseudowords), we used a meaning generation task, which required participants to retrieve the meaning of a vocabulary item from its form and is in many ways analogous to the experience of L2 readers having to consciously (even painstakingly) recall meanings of recently learned words. Furthermore, in the meaning generation task participants were required to retrieve word meanings without contextual support, probing a more advanced (abstracted) word knowledge. Importantly, however, learners' performance on a meaning generation task does not predict whether they might be able to access this lexical knowledge fluently under time pressure and, therefore, whether this knowledge is useful in real language use. To answer this question, tests of tacit knowledge were also used in this study, with the tacit knowledge conceptualized as quality of lexical representations. More specifically, two lexical decision tasks (LDTs) incorporating either form priming or semantic priming were deployed to test the quality of lexical representations established for critical L2 vocabulary from reading. A LDT is a widely established paradigm in L1 and L2 studies of visual and auditory lexical processing. In a speeded LDT, participants are presented with real and made-up words (nonwords) and are instructed to make a word/nonword decision for each item, as quickly and as accurately as possible. It is generally agreed that LDTs require participants to access lexical representations of word stimuli, at least when word-like nonword distractors are used (Joordens & Becker, 1997; Masson, 1995; Neely, 1991). In the present study, the LDT incorporated a form or semantic priming manipulation used to investigate more closely the corresponding aspects of tacit knowledge of critical vocabulary items. Priming involves the presentation of a related stimulus (prime) prior to another stimulus (target), such that it influences (often implicitly) the processing of the target, affecting response times (RTs) and accuracy levels for the target. A semantic priming study, for example, may investigate whether responses to the target word *doctor* are faster when it is preceded by the related word *nurse,* compared to the unrelated word *bread* (Meyer & Schvaneveldt, 1971).

Overall Method

Participants

Study participants were 48 advanced and high-intermediate adult L2 learners of English from diverse L1 backgrounds (see Appendix A), most of whom were undergraduate or graduate university students. They were recruited through local advertisements and received supermarket vouchers for their participation in the study.

Measures of Lexical Proficiency

Prior to reading the text, participants completed a vocabulary size test (VST) and a speeded LDT to estimate their L2 lexical proficiency. Participants' L2 vocabulary size (in word families) was estimated by calculating their VST scores (Nation & Beglar, 2007). Their fluency of access to L2 lexical representations (another measure of lexical proficiency) was evaluated using individual coefficients of variation of response latencies to words (WordCV) in a prereading LDT (Segalowitz & Segalowitz, 1993; see also Phillips, Segalowitz, O'Brien, & Yamasakia, 2004). The WordCV is calculated as an individual participant's standard deviation of RT divided by his/her mean RT and is interpreted as an indicator of the relative deployment of controlled and automatic processes in making lexical decisions (Segalowitz, 2000). Smaller WordCV values indicate a more automatized way of processing lexical information and reduced involvement of controlled processes. The pre-reading LDT, used to calculate WordCV values, contained 128 trials preceded by 16 practice trials. Both word ($n = 64$) and nonword ($n = 64$) stimuli were between 5 and 8 letters long ($M = 6.5$, $SD = 1.1$). The nonwords had on average 2.0 orthographic neighbours ($SD = 1.6$), and the words had on average 1.4 neighbours ($SD = 1.5$) and their frequency of occurrence per million (opm) (in CELEX), Baayen, Piepenbrock, & Gulikers, 1995 was 4.7 ($SD = 3.9$). Descriptive statistics for the VST and WordCV are provided in Appendix C.

Materials

The Text

Studies of incidental vocabulary learning from long texts have mainly used novels and short stories (Nation, 2006; Pellicer-Sánchez & Schmitt, 2010), but for adult L2 users outside of language learning programs the time available for recreational reading is often limited. L2 university students, for example, have to build their vocabularies while reading expository texts in their fields of study. For this reason, an introduction and four consecutive chapters from a nonfiction book *Freakonomics* by Levitt and Dubner (2006) were selected as the reading text in the present study. *Freakonomics* was number two on *The New York Times* best seller list in the nonfiction books category in 2006 and was likely to appeal to the study participants (this was confirmed in participants' ratings of text enjoyment in the postreading questionnaire). The book was unknown to the participants, as confirmed in the postreading interviews. *Freakonomics* is used as a supplementary textbook in some university courses in the United States and has an online *Student Guide* with comprehension questions, some of which were adapted for the comprehension test in the study. The length of the text used in the study was just over 40,000 words, with 95% lexical coverage provided by the first 7,000 most frequent English words (based on the British National Corpus; see Appendix B, Table B1).

Critical Vocabulary Items

For the purposes of this study, 48 words in the original text were replaced by 5–6 letter made-up words that were orthographically and phonologically similar to real English words (pseudowords; Appendix B, Table B2). Pseudowords were created by changing one letter in a real English word not related in form or meaning to the word it replaced in the text (e.g., the pseudoword *injent* that replaced the word *correlate* was created from a real word *inject*). Pseudowords were morphologically plausible; for example, *online* was replaced with *advern, legitimate* with *evotic*, *scholar* with *decop*, *violate* with *totate*, and so on. The pseudowords included 36 nouns, 6 verbs, and 6 adjectives/adverbs, with these grammatical classes corresponding to the words they replaced. Only content words that occurred at least six times in the text were replaced with pseudowords, and care was taken to avoid clustering of different pseudowords close together in the text, to maintain readability. Below is a passage from the text in which the verb *cheat* is replaced with the pseudoword *afuse*:

> But if a teacher *really* wanted to afuse—and make it worth her while—she might collect her students' answer sheets and, in the hour or so before turning them in to be read by an electronic scanner, erase the wrong answers and fill in correct ones.

Guessing From Context L1 Baseline Ratings

To estimate the degree of difficulty involved in guessing the meanings of the pseudowords from context, a rating procedure was conducted with 15 native speakers of English. Raters were presented with sentences from the text, in which critical words were deleted (replaced with gaps), and asked to (1) insert the missing word and (2) rate each instance on a 5-point scale (1 = *very easy*;

5 = *very difficult to guess*). The results confirmed that it was possible to infer the meanings of the missing words from context. On average, the raters found the task relatively easy ($M = 1.71$, $SD = 1.15$) and demonstrated good accuracy of responses ($M = .75$; $SD = .26$; 1 = correct, 0.5 = acceptable, 0 = incorrect). Although guessing word meaning from context may be easier in the L1, these results suggest that the level of contextual constraint was high enough for advanced L2 readers to infer word meanings from context.

The Learning Procedure

Study participants attended a prereading and a postreading (testing) session. In the individual prereading session they completed a language history and a topic knowledge/attitude questionnaire and two lexical proficiency tests (VST and LDT). At the end of the session, participants received the *Freakonomics* text, which they had to read within 10 days, keeping a reading log. In the log they had to record key points from the text, linking them with their personal experiences and prior knowledge. They received explicit instructions to focus on understanding and evaluating main ideas conveyed in the text and not to reread it. They also had to agree not to use dictionaries (which was confirmed in the postreading interview). Finally, participants were instructed to send an e-mail immediately after completing each chapter to request comprehension questions and to answer the questions as soon as possible before proceeding to the next chapter.

Measures of Comprehension

After completing each chapter of the text, participants received open-ended comprehension questions that were adapted from the *Student Guide to Freakonomics* (Appendix B, Table B3). These questions probed reorganizational understanding (Day & Park, 2005), as they required participants to go beyond recalling information, demonstrating their ability to understand and interpret complex ideas communicated by the authors (Ozuru, Briner, Kurby, & McNamara, 2013). Participants were instructed to respond to the questions without looking at the text. The accuracy of participants' responses to the questions was used as a measure of text comprehension. Responses were marked on a 4-point scale (3 = correct full answer, 2 = correct short answer, 1 = partially correct answer, 0 = incorrect answer) by two markers with postgraduate research degrees who had read the book and had access to the text during marking. Comprehension scores by question showed a high degree of agreement between the two markers, with inter-rater reliability of .92 (Solomon, 2004, based on the Spearman-Brown prediction formula). The resulting average comprehension score was 14.8 out of 24 points, that is, 62% comprehension. This result can be considered adequate based, for example, on a 55% adequate comprehension threshold in Laufer (1989) or a 56.5% comprehension threshold on a written recall test in Hu and Nation (2000). Both of these comprehension tests included open-ended or free recall questions that probed participants' understanding of main ideas and supporting details, thus making them comparable with the comprehension test used in this study.

The Testing Procedure

On completion of the reading task, participants returned for an individual postreading session, during which they completed a battery of tests measuring their knowledge of the critical pseudowords: first, two tests of tacit knowledge (i.e., an orthographic and semantic primed LDT, in that order); then a test of explicit knowledge (i.e., the meaning generation task). At the end of the session participants completed a postreading questionnaire that probed their attitudes to the text and use of vocabulary learning strategies (using a 5-point scale), followed by a short interview.

The Tests of Tacit Knowledge

To investigate whether lexicalization of the critical pseudowords had occurred (i.e., whether they were processed like real L2 words by the study participants), the critical items were used as form primes (stimuli close in form to L2 word targets) in a LDT. To investigate whether lexical semantic representations of contextually learned vocabulary had been established and integrated into existing L2 lexical-semantic networks of the participants, they completed a LDT, in which the newly learned pseudowords were used as semantic primes (stimuli related in meaning to L2 word targets). For the elicitation of tacit knowledge, primed LDT procedures were programmed and carried out in E-Prime (Psychological Software Tools, Inc., Pittsburgh, PA) and presented on a DELL PC (Intel® Core™2 Duo CPU) with a DELL LCD monitor (screen area: 1280 by 1024 pixels; refresh rate: 60 Hz). Detailed descriptions of the two tasks are provided below. In both tasks, the dependent variable was participants' RT on L2 lexical decisions.

The Test of Explicit Knowledge of Meaning

In the pen-and-paper meaning generation task, participants were presented with a randomized list of pseudowords from the text and asked to explain their meaning or provide a synonym. If they could not do this, participants were allowed to demonstrate their knowledge of meaning by using the pseudoword in a sentence or phrase. A score of 1 was awarded for correct or partially correct answers (e.g., for *detise*—meaning *incentive*—correct responses included "incentive," "something to make people do more good things and prevent people doing bad things"; for *cluff*—meaning *homicide*—partially correct responses were "violent crime," "cluff rate of urban young black people drop"). A score of 0 was awarded for incorrect answers or for no response.

Statistical Analyses

Linear mixed effects (lme) modeling was used in the analyses of the explicit and tacit knowledge, because its outcome can take account of the combination of fixed and random effects associated with learners and items. Indeed, in the present study, participants and items were included in all statistical models as crossed random effects. The lmer function (lme4 library package; Bates & Sarkar, 2010; Bates, 2011) of the interactive programming environment *R* was used in the data analyses.

A Generalized Linear Mixed Model (mixed logit model) was used to explore the effects of participant and item variables (explained below, see also Appendix C) on the binary score data from the meaning generation task measuring explicit knowledge of meaning. The statistical significance of the fixed effects was assessed using Wald's Z test (Jaeger, 2008; Wald, 1943).

In the lme analyses of the RT data from the primed LDTs, the difference in response latencies between the priming conditions (trials where primes were either orthographically or semantically related to the targets) and the control condition (trials where prime and target were unrelated) was taken to represent tacit knowledge (rather than absolute RTs). Thus the experimental condition (primed/nonprimed) was the primary-interest predictor in the statistical models fitted to the RT data, in both LDTs. All item and participant variables included in the statistical model for explicit knowledge were also included in the initial lme models fitted to the RT data, as secondary-interest predictors. Because these variables have been implicated in L2 word learning (as outlined in our review of the literature), their interactions with priming were also included in the initial models. In the regression analyses on the RT data, absolute t values larger than 2 indicated a significant effect at $p < .05$ (Baayen, Davidson, & Bates, 2008).

The RT analyses were conducted only on correct responses because interpretations of priming effects in lexical decisions rely on participants' ability to accurately recognize L2 word targets and because different cognitive processes are involved in making positive (accept) and negative (reject) responses (see Dijkstra, Miwa, Brummelhuis, Sappelli, & Baayen, 2010, for a similar approach in an lme analysis of RTs in lexical decisions). In addition, RTs on *Yes* and *No* responses maybe affected differently because the former are made with the dominant hand.

Due to the exploratory nature of the study (and following advice from a reviewer), the following modeling procedure was used in all data analyses (see also Barr, Levy, Scheepers, & Tily, 2013; Harrell, 2001; Jaeger, 2010; Jaeger & Kuperman, 2009). The predictor correlation matrix was scrutinized for highly correlated variables ($r > .2$), and necessary residualizations were performed (with variables representing secondary concepts regressed on those representing primary concepts), in order to reduce collinearity between predictors (Baayen, 2008; Jaeger, 2008, 2010). The full model with all theoretically and empirically motivated interactions was fit to the data. To avoid inflating Type 1 error rates, the initial full model included the maximal random effects structure justified by the data, that is, random slopes were fit for all fixed effects as appropriate (Baayen, 2008; Barr et al., 2013). Next, unreliable interactions, with $t < 1.00$, were removed, and the simplified model refit to the data. The full and simplified models were compared (at each simplification step) using the likelihood ratio test (Baayen et al., 2008), to verify that the simplified model was acceptable. After this, unreliable fixed effects with $t < 1.00$ (and corresponding random slopes) were removed (unless they were involved in reliable interactions) and the likelihood ratio test was used to verify that the new model fit was not worse. The constructed regression model was subjected to model criticism (Bayeen, 2008; Baayen & Milin, 2010), potentially harmful outliers (i.e., data points with absolute standardized residuals exceeding 2.5 standard deviations) were removed and the model was refit. Finally, the random effects matrix of the resulting model was inspected and effects that were highly correlated with intercept ($r = 1$) were removed (as redundant) to avoid model overparameterization (Baayen et al., 2008), and the likelihood ratio test was used again for model comparison.

Readers will find the output for the analytical steps outlined in this Overall Method section in the Supporting Information online.

Participant Variables

The participant variables initially included in the statistical models used in this study were: lexical proficiency (VST and WordCV), age, age of L2 acquisition (L2Age), gender, L1 group (alphabetic/nonalphabetic), and participants' ratings of their prior knowledge about economics (PriorKn; see Appendix C for descriptive statistics of the variables). Participants' gender (female $n = 38$) was used because the findings of psycholinguistic and neurological research show that males and females learn and process words differently, both in L1 and L2 (Chen et al., 2007; Dong et al., 2008; Kaushanskayaa, Marianb, & Yoo, 2011; Kimura, 1999; Ullman et al., 2002). The age of L2 acquisition was used because behavioural and neurological studies suggest that the architecture of L2 word representations is, at least to some degree, defined by the age of L2 acquisition (Kim, Relkin, Lee, & Hirsch, 1997; Silverberg & Samuel, 2004; Weber-Fox & Neville, 1999; Wuillemin & Richardson, 1994). The L1 may also affect the L2 reading ability and contextual L2 word learning of less proficient readers because of how different languages engage orthography, phonology, and semantics. The contrast between alphabetic and nonalphabetic languages is particularly striking; for instance, Chinese and English scripts require different visual processing mechanisms and differ in the way graphic units map onto linguistic units (Perfetti, Cao, & Booth, 2013). The following task- related variables were included in the initial model: individual reading comprehension scores (CompScore), self-reported word learning strategies (Strat), and ratings of enjoyment and interest in the text (Attitude). The use of vocabulary learning strategies was evaluated based on participants' responses to the following Likert-scale statements (1 = *strongly disagree*; 5 = *strongly agree*): (Q1) I ignored unknown words while reading the text; (Q2) I tried to guess the meaning of unknown words while reading the text; (Q3) I marked or noted down unknown words to look them up later. The final score was calculated as a sum of the three responses (with Q1 treated as a reverse score), because we were concerned with the use of strategies overall, rather than each of the specific strategies represented by these questions.

Item Variables

The concreteness/abstractness variable (based on the original words replaced by the pseudowords) was included in the statistical models because previous word learning studies show that it is easier to learn the meanings of words representing less abstract concepts (Crossley, Salsbury, & McNamara, 2009; Ellis & Beaton, 1993; Salsbury, Crossley, & McNamara, 2011). The concreteness value was obtained using hypernymy indices from Coh-Metrix (Graesser, McNamara, Louwerse, & Cai, 2004); for five items (adjectives/adverbs) not in Coh-Metrix this was set to 1 (as the hypernymy index for the least concrete words in Coh-Metrix is 1). The replaced letter position (initial, final, other) was also included in the model as a dummy variable (Letter).

In addition, the following text-specific item variables were included in the initial model: rate of occurrence in the text (RateOccur), ease of guessing from context rating (GuessRTG), spacing of items occurrences across chapters (Range), and Keyness calculated using a keyword extractor tool (www.lextutor.ca/keywords/) for the original words replaced by pseudowords. The keyword tool identifies defining (salient or prominent) lexis in a specific text or corpus by comparing frequency in the text to frequency in a reference corpus (i.e., the Brown University corpus of American English), on a per-million basis. The keyness factor is calculated for words that are at least 10 times more numerous in a given text than in the Brown reference corpus. Critical items that were not assigned a keyness factor (i.e., below the threshold) were assigned the factor of 1 because their use in the text was considered to be broadly aligned with that in the Brown corpus. Keyness is interpreted here as an objective measure of a word's salience in the text: the greater the keyness index, the more salient a word is likely to be in a given text (Bondi, 2010).

Transformation and Standardization of Variables

The participant and item variables were examined to establish if they needed to be transformed to bring the variables closer to normal distribution. This resulted in the following transformations that were used in all analyses: strategy use, item keyness, and ratings of ease of guessing from context were logarithmically transformed, and rate of occurrence was inverse transformed (invOccur = −1,000/Occur). VST data was logarithmically transformed because the data were large whole numbers and covered a range of values. Next, all nondichotomous variables were standardized and centered using the scale() function in R to avoid multicollinearity (Belsley, Kuh, & Welsch, 1980) and to enable easier interpretation of the results.

In addition to the transformations of item and participant variables described above, additional variables specific to the primed LDTs were inspected prior to the RT data analyses, and the following transformations were made to bring the variables closer to normal distribution: the RT data were inverse transformed, scaled by 1,000, and multiplied by −1 (RTinv = –1,000/RT) (Dijkstra, Miwa, Brummelhuis, Sappelli, & Baayen, 2010); frequency of word targets (HAL), the number of phonological and orthographic neighbors of the target (obtained from the *English Lexicon Project* website; Balota et al., 2007) were log transformed. Trial number and participant RT on a preceding trial (PrevRT) were included in the models as longitudinal predictors (Baayen & Milin, 2010). All additional predictors were standardized and centered.

We begin our presentation of results with the meaning generation test because more is known from previous research about the effects of participant and item variables on the acquisition of explicit L2 vocabulary knowledge. We then detail the design of the two primed LDTs used to measure tacit knowledge of the contextually learned vocabulary and report on the results of the form and semantic priming task, in turn.

Explicit Knowledge Results

The average score on the meaning generation task was .2 ($SD = .11$), that is, on average, participants learned the meanings for about 10 of the 48 items. This result is comparable to previously reported results for incidental learning of word meanings from context, for example, 15% in the study by Brown et al. (2008) for the reading only group; 18% in Waring and Takaki (2003) (measured by the meaning-translation posttest in both cases), and 14% gain in the meaning recall test in Pellicer-Sanchez and Schmitt (2010).

The data set comprised 2,304 responses (510 correct). An initial mixed logit model was fit to the response accuracy data from the meaning generation task, following the modeling procedure specified in the Overall Method section. The following interactions were included in the initial model: (1) between rate of occurrence and lexical proficiency (VST and WordCV), based on the findings in L2 and L1 studies that the effect of frequency of encounters with a word in reading may be modulated by proficiency (e.g., Frishkoff et al., 2008, Zahar et al., 2001); (2) between rate of occurrence and text comprehension, based on the finding that text comprehension affects contextual learning of word meanings, when rate of occurrence is controlled for (Pulido, 2007); and (3) between the spacing of item repetitions across chapters and lexical proficiency, based on the finding by Frishkoff et al. (2008) of better learning in the spaced than in the massed condition, for high (but not for low) verbal proficiency participants.

After inspecting the correlation matrix of the fixed effects in the initial model (see Table S1 in the Supporting Information online), highly correlated variables were decorrelated, and the obtained residuals were taken as new, orthogonalized predictors. All residualized predictors correlated well with the original variables (see r values in parentheses below), confirming that the decorrelations did not change the nature of the original predictors. Keyness was regressed on rate of occurrence ($r = .90$), range was regressed on rate of occurrence and keyness ($r = .94$), ease of item guessing was regressed on rate of occurrence and item concreteness, ($r = .84$), age of L2 acquisition was regressed on age ($r = .94$), VST was regressed on WordCV ($r = .94$), comprehension scores were regressed on VST and WordCV ($r = .80$), use of vocabulary learning strategies was regressed on VST ($r = .99$), prior knowledge was regressed on age and VST ($r = .96$), and item concreteness was regressed on rate of occurrence and range ($r = .97$). A summary of results for the final simplified model is reported Appendix D. For the initial full model, see Table S2 in the Supporting Information online.

There were significant simple effects of lexical proficiency, text comprehension, rate of occurrence, and item spacing across chapters on participants' ability to correctly retrieve the meanings of critical items. However, because these predictors were also involved in reliable interactions, the results of the interactions will be reported first.

The following interactions included in the model were shown to be statistically reliable: (1) between rate of occurrence of the pseudowords in the text and participants' comprehension scores ($z = 3.63$, $p < .001$), (2) between rate of occurrence and participants' L2 lexical proficiency – VST ($z = 2.21$, $p < .05$), and (3) between participants' lexical proficiency and the spacing of items across chapters ($z = –2.05$, $p < .05$). The interaction between rate of occurrence and lexical fluency (WordCV) was in the expected direction but did not reach significance ($z = 1.34$, $p = .18$). These interactions are depicted in Figure 1 (this and other plots are based on model predictions).

The positive effect of rate of occurrence on the learning of meaning was strongly modulated by participants' ability to understand the text, that is, readers with higher comprehension scores were better able to take advantage of multiple occurrences. This is shown in Figure 1a. A ceiling effect was observed for the best comprehenders around 25 occurrences. The effect of rate of occurrence was also modulated by participants' L2 lexical proficiency, as shown in Figure 1b: More advanced participants were likely to learn the meanings of critical items after fewer encounters compared to less advanced participants.

The interaction between participants' L2 lexical proficiency and the spacing of items across chapters showed an interesting trend, shown in Figure 1c. Closer spacing of repetitions (within the same chapter) was, overall, more beneficial than broader dispersion across chapters; but, at higher L2 proficiencies, the drop in learning occurred more gradually while, at lower proficiencies, little learning was observed unless an item was encountered within the same chapter. This finding is considered further in the discussion section below.

The following main effects were also observed in the analysis. Contextual word learning progressed faster for readers who reported greater use of vocabulary learning strategies ($z = 4.08$, $p < .001$) and those who enjoyed reading the text more ($z = 4.14$, $p < .001$). Females learned more than males ($z = 2.54$, $p < .05$). Furthermore, study participants were more likely to correctly retrieve the meanings of the critical items that had a higher keyness index ($z = 3.95$, $p < .001$) and that denoted more concrete concepts ($z = 2.72$, $p < .01$).

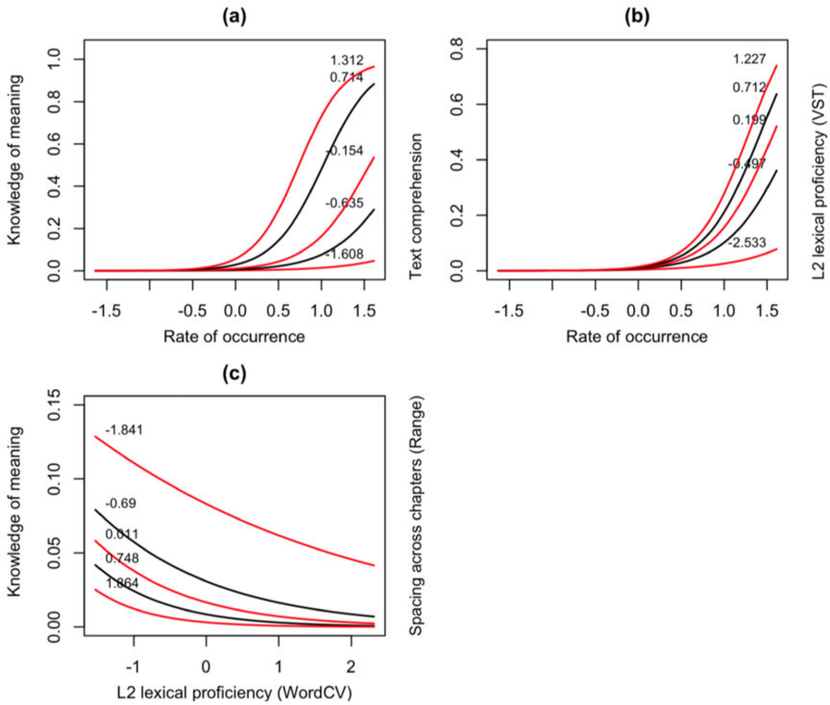

Figure 1 Explicit knowledge of meaning. Partial effects plots showing interactions between (a) rate of occurrence and participant text comprehension ($z = 3.63$, $p < .001$); (b) rate of occurrence and participant VST ($z = 2.21$, $p < .05$); (c) participant WordCV and spacing of item occurrences across chapters (Range, $z = -2.05$, $p < .05$).
Note 1: Red and black lines represent quartiles (0% – 25% – 50% – 75% – 100%) of the interaction variable: (a) comprehension, (b) L2 vocabulary size, (c) range across chapters.
Note 2: Lower CV values represent higher L2 lexical proficiency.

Discussion of Explicit Knowledge Results

The results of the explicit knowledge of meaning test showed modest vocabulary gains for the novel vocabulary items encountered while reading a long connected L2 text. As predicted, these gains were affected by item and learner variables. It is not surprising, in view of previous findings, that the largest effect size (based on the standardized coefficients: $\beta = 2.90$, $SE(\beta) = 0.42$, Appendix D) was observed for rate of occurrence. Interestingly, higher occurrences were not equally beneficial to all readers. Even when the critical items occurred multiple times (up to 88), readers who did not understand the main ideas of the text could not retrieve the meanings of the items. Further, the lower the learner's lexical proficiency, the more contextual encounters with a new L2 word were necessary for the learning to occur. The proficiency advantage was particularly prominent between 12 and 25 occurrences in our study, with little explicit learning for items for which there were fewer than 12 encounters, regardless of reader lexical proficiency. Vocabulary meaning gains were also reliably affected by the items' salience in the text (indexed by keyness); presumably, words carrying key ideas and concepts conveyed in the text were more likely to attract readers' attention, leading to greater learning gains (Schmidt, 2001).

Another attention-related predictor—the use of vocabulary learning strategies—also had a reliable positive effect on learning. Learners who reported using vocabulary strategies (e.g., noticing, underlining, trying to work out the meaning of unknown words) were more likely to have higher gains in explicit knowledge of meaning. This is despite explicit instructions to focus on the meaning of the text and not to look up unknown words. This finding suggests that some minimal use of vocabulary strategies is critical for creating explicit associations between form and meaning. Additionally, the reported level of interest and enjoyment had a positive effect on learning in its own right, highlighting the value of a "good read" in L2 learning.

Finally, similar to Frishkoff et al. (2008), we found an effect of the spacing of repetitions that interacted with proficiency. However, our results appear to be orthogonal to those in Frishkoff et al.'s L1 word learning, in which the use of spaced practice was more effective than massed practice, for more skilled (but not less skilled) learners. In the present study, all readers learned more when a new item was encountered in the same chapter than when it was encountered across multiple chapters and, at lower proficiencies, this made the difference between some learning and no learning. At higher proficiencies, increasing range resulted in a more gradual decrease in the acquisition of explicit knowledge. Although this finding may seem unexpected, a closer comparison of the tasks and the conceptualization of spacing in the two studies provides a plausible explanation of the differences. First, Frishkoff et al. (2008, p. 913) operationalized the narrow-spaced (massed) condition as 3–5 trials apart and the wide-spaced condition as 14–25 trials apart. In our study, the spacing within the same chapter fits both of these conditions, as in some cases the critical items occurred within the same paragraph or page and, in others, on different pages of the chapter. Individual chapters were read mostly within 1 day, but there was often a short break (1–2 days) between chapters (based on the reading journals). Critically, in their deliberate learning study, Frishkoff et al. asked participants to generate the meanings of new words after each contextual encounter. Such deliberate vocabulary learning, most likely, involves extensive use of metacognitive strategies, as pointed out by the authors. In our study, with no emphasis on vocabulary learning, participants were unlikely to ever attempt to articulate definitions for the critical items explicitly.

The benefits of more closely spaced encounters in the present study may be interpreted in reference to the instance-based framework of word learning (Bolger, Balass, Landen, & Perfetti, 2008; Hintzman, 1986). In this framework, episodic memory traces are established initially for contextually learned words, and these episodic representations tend to decay over time. After multiple encounters, the core meaning of a word is abstracted (becomes independent from context), that is, a lexical semantic representation for the word is established and integrated into the lexical semantic memory of the learner. In the present study, encounters within the same chapter may have attenuated the decay of the episodic memory traces, allowing more time for the lexical semantic representations to develop and the abstraction of the core meaning to take place. This should be particularly beneficial for less proficient learners, whose L2 lexical semantic networks are underdeveloped and for whom, therefore, it takes longer to learn new words from context. The conjecture that the developmental state of the learner's mental lexicon modulates the trajectory of contextual L2 lexical development is further tested in the tasks probing participants' tacit knowledge of the critical items, described below in turn.

Tacit Knowledge: Form Priming

To evaluate the quality of lexical representations established for the pseudowords, the same participants completed a speeded LDT, in which the pseudowords encountered in the text were used as orthographic form primes. This task was based on the masked form-priming task described in Davis and Lupker (2006, Experiment 1) that revealed a Prime Lexicality Effect (Forster & Vereš, 1998) with L1 stimuli. Forster and Vereš (1998) observed that, in a LDT, when a word target (e.g., *FUNCTION*) is preceded by a form-related nonword prime one letter different from that target (e.g., *bunction*), the target is recognized faster than when it is preceded by an orthographically unrelated prime (e.g., *mushroom*). This effect is presumed to be due to the sublexical letter-level facilitation of the target by the prime. This facilitation effect is attenuated if the orthographically related prime is a word (e.g., *junction*) because of the competition between the lexical representations of the prime and the target. Because the effect was modulated by the lexical status of the prime (word vs. nonword), it became known as the Prime Lexicality Effect (PLE).

Forster and Vereš (1998) were able to observe the PLE only under unmasked priming conditions (when both primes and targets were presented for 500 milliseconds), but Davis and Lupker (2006) reported that the PLE also occurred under automatic priming conditions. Using shorter stimuli, they observed the PLE with primes that were masked and presented for 57 milliseconds. Furthermore, the PLE observed by Davis and Lupker was even more pronounced; similar to Forster and Vereš (1998), facilitation was recorded for pairs with orthographically related nonword primes but, instead of "no effect," their experiments generated inhibition with orthographically related word primes. In L2, Elgort (2011) argued that the PLE can be used as an indicator of word learning in vocabulary acquisition studies, because it distinguishes between words and nonwords, as experienced by the learner. In Elgort's intentional L2 word learning study, the PLE was observed for known L2 words and for deliberately learned L2 words, under unmasked priming conditions.

Method

Stimuli

Experimental stimuli for the form-priming task included 48 word targets paired with related nonword primes, related pseudoword primes or unrelated word primes; 48 nonword targets paired with related nonword primes, related word primes or unrelated word primes; and 32 unrelated filler trials with word and nonword targets. All stimuli were 5 or 6 letters long. The average frequency of the word targets was 4.6 occurrences per million (opm; CELEX) and their average orthographic neighbourhood size (*N*; Coltheart, Davelaar, Jonasson, & Besner, 1977) was 1.5. For the pseudoword primes, the average *N* was 2.9 and for the nonword primes it was 2.7. The related nonword primes used on critical trials were constructed by changing the same letter of the target word as in the pseudowords (e.g., the nonword *aguse* was created by changing the second letter of the target word *AMUSE*, which had a corresponding pseudoword *afuse*, meaning *cheat*). The average number of shared neighbours between pseudoword primes and word targets and between nonword primes and word targets was 1.1. However, having encountered the pseudowords in the text, participants were likely to have lexicalized some of them, adding a shared neighbour to the corresponding nonword–word pairs.

Design

On critical trials word targets were preceded by (1) orthographically related pseudoword primes from the text [*flane* (meaning *eliminate*)—*FLAKE*]; (2) orthographically related nonword primes never encountered by participants prior to the experiment [*flave*—*FLAKE*]; or (3) unrelated words (control primes), that is, words unrelated to the target in their spelling or meaning [*drown*—*FLAKE*] (Appendix E, Table E1). Nonword targets in the experiment were preceded by (1) orthographically related nonword primes [*chesk*—*CHESH*], (2) orthographically related word primes [*chess*—*CHESH*], or (3) control words unrelated to the target [*rapid*—*CHESH*]. On filler trials, 16 word targets were preceded by unrelated nonword primes [*erame*—*ABBEY*] and 16 nonword targets were preceded by unrelated nonword primes [*blorm*—*MURCH*]. Primes were always presented in lowercase and targets in uppercase. The nonword trials or the fillers were excluded from the analysis as only the critical trials are relevant to addressing the research questions.

For counterbalancing, three presentation lists were created, with each critical word target presented in one of the three experimental conditions in each list, and all targets presented in all three conditions across the three lists. The same counterbalancing approach was used for the 48 nonword targets, while the filler trials were the same in all lists. Each list contained 128 pairs, arranged in a pseudo-random order that was kept the same for all lists. The three lists were rotated across three groups of participants, based on the order of their participation.

Procedure

The masked priming procedure from Davis and Lupker (2006) was used in the task. Each trial consisted of the following events: presentations of a forward mask represented by 5 or 6 hash signs (#) for 500 milliseconds, immediately followed by the prime presented for 56 milliseconds, immediately followed by the target presented for 500 milliseconds (Appendix E, Figure E1). The response deadline was 2,500 milliseconds. All stimuli were presented in the same position in the middle of the screen. The study participants were instructed to decide, as quickly and as accurately as possible, whether the uppercase stimuli (targets) were English words or not, using the *Yes* and *No* buttons on a button response box connected to the computer. They used the index finger of their dominant hand for *Yes* responses and nondominant hand for *No* responses. At the start of the procedure, participants completed a practice block of 21 trials, during which they received immediate feedback on the accuracy and speed of their responses.

Results

The primary question under examination here was whether positive form priming is attenuated when the critical pseudowords (encountered by participants in

Table 1 Mean response times (in milliseconds), accuracy, and standard deviations for correct responses to word targets in the form-priming experiment, by condition

Condition	ACC	*SD*	RT	*SD*	Diff
Control [*drown—FLAKE*]	.91	.28	910	447	NA
Pseudoword priming [*flane* (*eliminate*)—*FLAKE*]	.91	.29	907	437	−3
Nonword priming [*flave—FLAKE*]	.89	.32	887	418	−23

reading) are used as form primes and whether pseudoword priming is affected by the participant and item predictors. In addition, nonword priming was examined to confirm that reliable facilitative priming was present with the same participants and for the same targets when form primes did not have a lexical status (were word-like letter strings with no meaning). After removing trials with incorrect responses, the final data set consisted of 2,074 observations for word targets. The average accuracy of responses to word targets in the priming data was 90% (see Table 1).

The experimental data were split into two sets (pseudoword and nonword), and separate lme models were fit to each data set. This is because item variables associated with the reading task (i.e., rate of occurrence, keyness, range, ease of guessing, degree of concreteness of the base word, comprehension scores, prior knowledge of the topic, participants' attitudes to the text) were only relevant in the pseudoword-priming analysis and were irrelevant in the nonword-priming analysis (and any effects of these predictors present in the nonword-priming analysis would have been difficult to interpret and would, most likely, be present by chance).[1]

Results of the form-priming analysis for the pseudoword set are presented in Appendix F, Table F1. For the initial full model, see Table S3 in the Supporting Information online. The simple effect of form priming with critical pseudoword primes [*flane* (meaning *eliminate*)–*FLAKE* compared to *drown–FLAKE*] had a trend toward significance ($t = -1.84$, $SE = .01$). Importantly, form priming interacted reliably with age of L2 acquisition ($t = -3.13$, $SE = .02$); participants who started learning English earlier in life were more likely to give slower responses in the form-priming condition—an indication that L2 lexical representations of the pseudowords were established for these learners (Figure 2a). In addition, an interaction between priming and pseudoword concreteness (Figure 2b) showed that pseudowords that denoted more concrete concepts were more likely to be lexicalized than those that denoted more abstract

(a) an interaction between priming and participant age of L2 acquisition (t=-3.13 SE=.017)

(b) an interaction between priming and pseudoword concreteness (t=1.76 SE=.013)

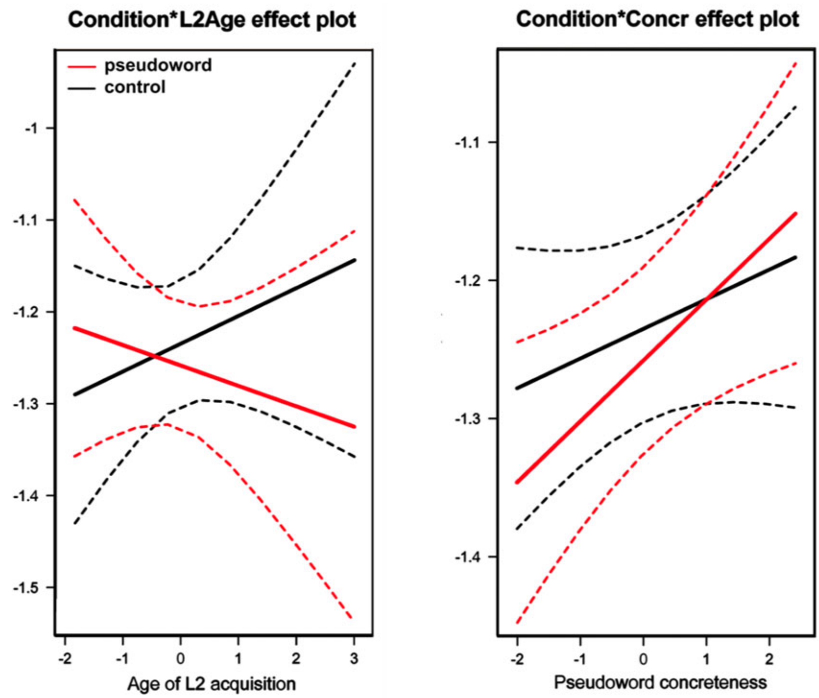

Figure 2 Form-priming with pseudoword primes. Partial effects plots (with 0.95 confidence limits based on the standard-normal distribution) showing interaction between priming and predictors.

concepts, but this interaction did not reach statistical signifiance ($t = 1.76$, $SE =$ .01).

The nonword form-priming results can be found in Appendix F, Table F2; for the initial full model, see Table S4 in the Supporting Information online. Nonword form priming [*flave–FLAKE* compared to *drown–FLAKE*] was statistically reliable ($t = –3.79$, $SE = .01$).

Discussion

The results of the form-priming task suggest that reading a single authentic text for meaning may not be enough for the reliable lexicalization of previously unknown L2 words to occur. Although facilitation was attenuated when critical pseudowords were used as form primes (i.e., facilitation was numerically smaller and not as reliable as that with nonword primes), this facilitation was larger than what is expected in form priming with known word primes (cf. Davis & Lupker, 2006).[2] There are two likely reasons for the observed outcome: (1) robust representations may have been established for some (but not all) critical items and (2) robust lexical knowledge may have been acquired by some (but not all) learners. Indeed, our results show that contextual word learning was more effective for L2 speakers who started learning English in their early childhood and that chances of robust learning were higher for more concrete items. Because previous research has shown that inhibition in L1 form priming is more likely to be observed for more skilled readers and spellers whose lexical representations are more precise (Andrews & Hersch, 2010; Andrews & Lo, 2012), it is also possible that a weaker attenuation of form priming observed for the participants who started learning English at a later age, in the present study, is due to their overall less-developed and less fine-tuned L2 mental lexicons.

Tacit Knowledge: Semantic Priming

The quality of lexical semantic representations of the pseudowords was examined using a semantic priming task, the design of which was based on McRae and Boisvert (1998, Experiment 1). In positive semantic priming, a word target is recognized faster when it is preceded by a word prime related to it in meaning, compared to the control condition—when the same target is preceded by an unrelated word prime (Meyer & Schvaneveldt, 1971). This facilitation effect is taken to reflect the properties of lexical semantic knowledge, where semantically related lexical representations are interlinked.

Method

Stimuli and Design

Three experimental conditions were constructed for the semantic priming task: (1) related pseudoword–word, (2) related word–word, and (3) unrelated word–word (control condition). The related word–word condition was included in the design to verify that semantic priming occurred with real word primes that were presumed to be familiar to the participants and had the same or very similar semantic relationships with the targets as the critical items. In our study, semantic relatedness between the prime and the target was realized either as a semantic feature overlap (e.g., *quote–cite*; *expert–amateur*) or as the likelihood of their cooccurrence in the language (e.g., *candidate–electing*; *online–dating*).

Critical pseudoword primes ($n = 48$) were matched with 48 related word targets (e.g., *trimp* [meaning *opponent*]—*challenger*; *grude* [meaning *candidate*]—*electing*). The average length of the targets was 8.3 letters ($SD = 1.9$), their average word frequency was 17.9 opm ($SD = 19.9$). In addition, the same targets were matched with real-word primes that were the same (or very close) in meaning to the pseudowords (e.g., *opponent*—*challenger*; *candidate*—*electing*). Finally, 48 words unrelated in meaning or form to the targets were used to construct the control condition (e.g., *nylon*—*challenger*; *speck*—*electing*). These control primes were also selected from the *Freakonomics* text, in order to match them with the pseudowords in terms of the recency of encounter and topic familiarity. The average length of the pseudoword primes was 6 letters ($SD = 1.25$), because inflected forms of pseudowords from the text were used; the average length of related word primes was 6.9 letters ($SD = 1.4$), their average frequency was 21.2 ($SD = 19.2$); and the average length of unrelated word primes was 6 letters ($SD = 1.3$), their average frequency was 9.2 ($SD = 7.3$). To confirm that the related and unrelated conditions were different from each other, similarity values were obtained using the *Latent Semantic Analysis* (LSA) online tool (http://lsa.colorado.edu) that represents meaning similarity statistically (based on distributional characteristics of words in a large body of text) (Landauer, Foltz, & Laham, 1998). LSA similarity scores have been shown to closely match those of human similarity judgments. The results for the two conditions (related word–word and unrelated word–word) were significantly different from each other: $F_{(1,47)} = 99.79$, $p < .001$ ($\eta_p{}^2 = .680$), with the mean score of .28 ($SE = .02$) for the related condition and .05 ($SE = .01$)[3] for the unrelated condition.

In addition to the critical stimuli, each experimental list included 32 unrelated filler words (to equalize the number of related and unrelated word pairs) and 128 unrelated nonwords (for the LDT). Nonwords were constructed by replacing one letter in a real English word and were orthographically and phonologically plausible. Their average length was 6.5 letters ($SD = 1.3$). The fillers and nonwords were excluded from the statistical analysis because they were irrelevant to the research questions.

Three counterbalanced experimental lists were constructed in such a way that critical word targets appeared only once per list in one of the three experimental conditions and all targets appeared in all three conditions across all lists. Each list included 256 individual trials, with the targets presented in the same pseudo-random order in the three lists. The three lists were rotated across three groups of participants, based on the order of their participation. Filler stimuli were the same in all lists. To avoid the use of strategies by participants, primes and targets were not explicitly paired, but presented as adjacent items in a longer list of stimuli, with lexical decisions made on each stimulus (see Appendix E, Table E2 and Figure E2).

Procedure

Participants first completed a practice block of 32 trials, during which they received immediate feedback on the accuracy and speed of their responses. The procedure started with a presentation of an asterisk (*) in the middle of the screen for 500 milliseconds, which was replaced by the stimulus presented until response, followed by a blank screen presented for 200 milliseconds, and immediately replaced by the next stimulus from the experimental list. No response deadline was used in this procedure but, based on the distribution of the RT data, only responses between 200 milliseconds and 4,000 milliseconds were included in the analysis. Participants were instructed to decide, as quickly and as accurately as possible, whether the displayed stimulus was an English word or not. They registered their responses by pressing *Yes* or *No* on the response box, using the index finger of the dominant hand for *Yes* and nondominant hand for *No*.

Results

The goal of this semantic priming task was to evaluate the quality of lexical semantic representations established for the contextually learned critical vocabulary. Facilitative semantic priming in lexical decisions is observed if (a) participants can access lexical semantic representations of the experimental stimuli fluently (under time pressure) when they are visually presented and (b) primes and targets in the related (priming) condition are processed by participants as semantically similar (i.e., lexical semantic representations of the targets are activated on presentation of semantically related primes). Semantic facilitation observed with the critical pseudoword primes would indicate that robust lexical semantic representations of the pseudowords had been established and integrated into the L2 lexical semantic networks of the readers. After removing trials with incorrect responses and four word targets (*robber*, *intolerance*, *territory*, and *indicate*) with low response accuracy (<60%), the final dataset consisted of 6,648 observations for word targets. The average accuracy of responses to word targets in the data was 97% (Table 2).

The experimental data were split into two sets (pseudoword and word), and separate lme models were fit to each dataset. Separate analyses of the pseudoword and word priming were motivated by the same considerations as in the form-priming task, that is, the relevance of the item variables associated with the prior reading of the text.[4]

Table 2 Mean response times (in milliseconds), accuracy, and standard deviations for correct responses to word targets in the semantic-priming experiment, by condition

Condition	ACC	*SD*	RT	*SD*	Diff
Control [*frown* — *contestant*]	.95	.21	850	335	NA
Pseudoword priming [*flane* (*eliminate*)—*contestant*]	.95	.21	862	368	12
Word priming [*eliminate* — *contestant*]	.97	.17	807	320	−43

Results of the semantic priming analysis for the pseudoword set are presented in Appendix G, Table G1 (for the initial full model, see Table S5 in the Supporting Information online). The simple effect of semantic priming with critical pseudoword primes was not reliable ($t = –1.24$, $SE = .02$). Importantly, semantic priming interacted with three participant variables and one item variable (as shown in Figure 3); it also interacted reliably with participant response latencies to primes. Semantic facilitation was reliably more likely to occur for participants who started learning English earlier in life ($t = 2.40$, $SE = .01$; Figure 3a). Facilitation was also predicted for those who scored higher on the text comprehension questions ($t = –1.91$, $SE = .01$; Figure 3b) and those who reported using vocabulary learning strategies ($t = –1.92$, $SE = .01$; Figure 3c); as well as for the items with a higher keyness factor ($t = –1.61$, $SE = .01$; Figure 3d), but the latter interaction was not statistically reliable. Interestingly, semantic priming was positive when responses to primes were slower, while inhibition was predicted for faster responses ($t =$ -4.18, $SE = .01$).

Results of the semantic priming analysis for the word set can be found in Appendix G, Table G2 (and for the initial full mode, see Table S6 in the Supporting Information online). This analysis revealed reliable semantic priming on the related word–word condition ($t = –2.06$, $SE = .01$).

Discussion

Although, overall, only weak lexical semantic representations were established for the vocabulary items learned incidentally from reading, a number of participant and text variables affected contextual acquisition of meaning. Similar to the form-priming findings, semantic priming was more likely to occur for L2 readers who began learning English in early childhood, suggesting that age of acquisition is an important factor in contextual L2 word learning. Better comprehenders were better at contextual word learning (cf. McKeown, 1985) as were readers who used vocabulary learning strategies. Meaning acquisition was also positively affected by item salience (keyness) in the text.

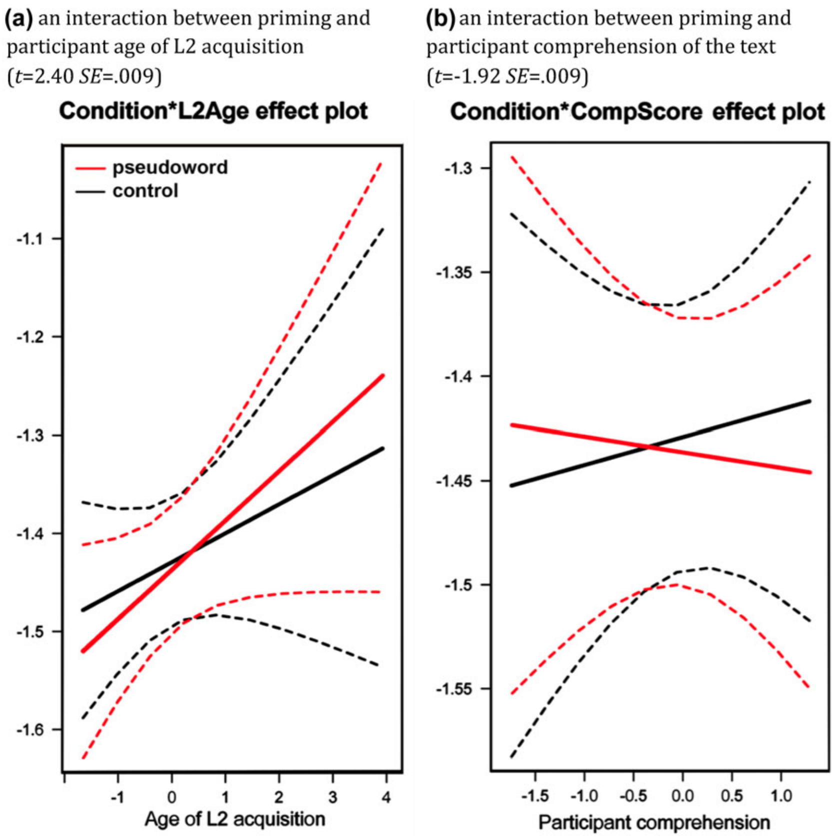

Figure 3 Semantic priming with pseudoword primes. Partial effects plots (with 0.95 confidence limits based on the standard-normal distribution) showing interaction between priming and predictors.

Semantic priming was less likely to be observed on faster responses to primes. The direction of this interaction is most likely related to the fact that the present study concerns early stages of incidental vocabulary learning. Because the LDT does not require deep semantic processing, some minimal level of activation of a word's lexical representation may be enough to make a decision about its lexicality. Semantic priming, on the other hand, requires participants to process lexical semantic features of the prime to such a degree that they preactivate the overlapping aspects of the lexical semantic representations of the targets. Presumably, the weak lexical semantic representations of

(c) an interaction between priming and participant use of vocabulary learning strategies (t=-1.91 SE=.011)

(d) an interaction between priming and item keyness (t=-1.61 SE=.009)

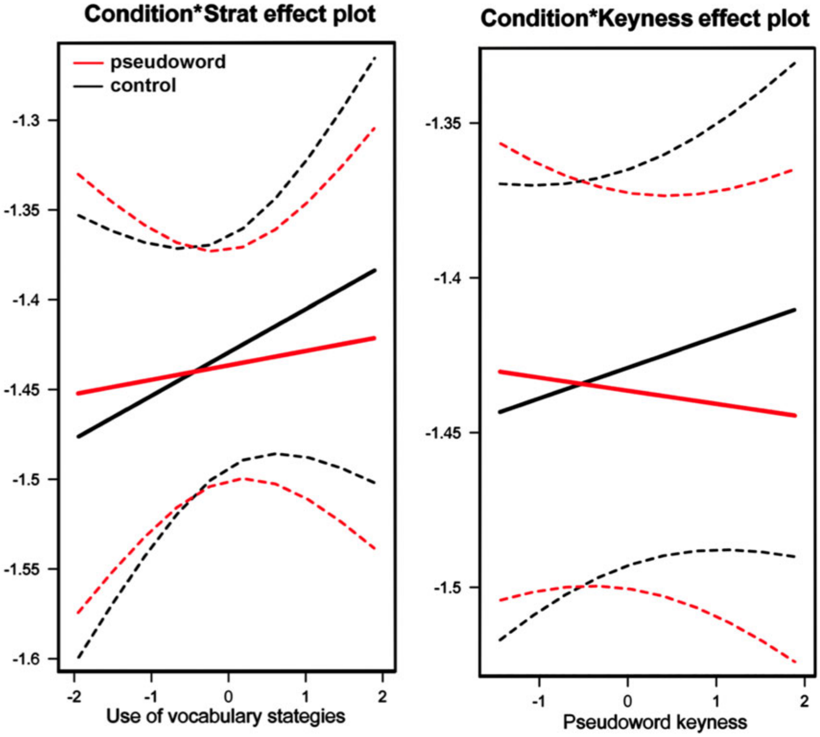

Figure 3 (*continued*)

the critical pseudowords need longer processing times to be fully activated than representations of known words, processed automatically. Therefore, semantic priming in the early stages of word learning is more likely to be observed when responses to primes are slower.

Quality of lexical semantic representations of the primes is also at the core of another potential explanation of this interaction. In an L1 study requiring participants to fully identify newly learned word primes, Dagenbach, Carr, and Barnhardt (1990) reported that weakly acquired words (i.e., words that participants could recognize in a multiple-choice task, but could not recall in a cued recall task) inhibited the recognition of semantically related word targets, while word primes that participants could both recognize and retrieve facilitated the recognition of related targets. To account for this result, Dagenbach et al. (1990) suggest that, in the course of identifying weakly known word primes, semantically related targets are temporarily suppressed as their competing semantic neighbours. This conjecture finds support in the present study, that is, inhibition (rather than facilitation) was observed for late L2 learners (Figure 3a). Thus, the observed interaction between the polarity of priming and response latencies to primes may be related to the stimulus onset asynchronies (SOAs) between primes and targets, as faster responses to primes resulted in shorter SOAs, that is, less time for the temporary suppression of the related targets to dissipate, hence the inhibition effect.

General Discussion

This study investigated gains in explicit and tacit lexical knowledge of novel vocabulary encountered while reading a long connected L2 text and participant and item variables that affect contextual L2 word learning. Inevitably, fewer learning variables can be controlled in studies conducted under naturalistic learning conditions (cf. Pellicer-Sanchez & Schmitt, 2010), compared to experimental laboratory studies. Although we endeavored to minimize the effect of these variables by providing clear task instructions and verifying participants' behavior by examining their logs and debriefing them in the second session, we acknowledge that there is a limit to what can be effectively controlled in this context (e.g., the degree of confidence that they did not use dictionaries or did not refer back to the text when responding to comprehension questions). However, we would argue that observing L2 contextual word learning under naturalistic conditions extends our knowledge about the multifaceted process of L2 lexical development from reading.

The results show that, at these relatively early stages of contextual word learning, participants' gains in the explicit knowledge of meaning were positively affected by their L2 lexical proficiency and their use of vocabulary learning strategies, such as noticing new words and attempting to guess their meanings. Larger gains were revealed for better comprehenders and readers who reported higher interest and enjoyment. More concrete L2 vocabulary items were easier to learn. Contextual word learning was also reliably affected by the number of times an item was encountered (rate of occurrence) and by item salience (keyness) in the text. The learning advantage associated with rate of occurrence was modulated by readers' text comprehension and their L2 lexical proficiency (Figure 1a & b). Participants whose text comprehension was low were less likely to learn the meanings of the new vocabulary items even if they occurred multiple times in the text. The number of encounters with a word needed for learning was higher for less proficient participants. Finally, at lower proficiencies, contextual learning was successful only when items occurred within the same chapter. At higher proficiencies, same-chapter distribution was also more effective, but the drop in learning outcomes observed for broader spacing across chapters was more gradual and less dramatic (Figure 1c).

Having conceptualized tacit lexical knowledge as quality of lexical representations, we examined the formal and semantic aspects of lexical representations established for the novel L2 vocabulary after reading the text. Our results demonstrate that contextual word learning is a slow and incremental process. Encountering unknown L2 vocabulary items during reading triggers the lexicalization process, but falls short of establishing robust formal-lexical or lexical semantic representations for these items, at least in the early stages of learning. The development of tacit knowledge was influenced by fewer participant and item variables, and only some of these variables were the same as those that influenced the development of explicit knowledge. Similar to the meaning generation task, better comprehenders and those who used vocabulary learning strategies demonstrated higher tacit knowledge of meaning, and item concreteness and keyness had a positive effect on the establishment of forma-lexical and lexical semantic representations of the contextually learned words, respectively. Unlike explicit knowledge, however, tacit knowledge was reliably influenced by the age when participants started learning the L2, with faster and more robust lexicalization of contextually encountered vocabulary observed for those who started learning English earlier in life. Contributing to the debate in the L2 development literature about the relative contribution of L2 proficiency and age of L2 acquisition to L2 learning/acquisition (Birdsong, 2006; Perani et al., 1998), the present study reveals that lexical proficiency plays an important role in the development of explicit knowledge, while age of L2 acquisition affects tacit knowledge of contextually learned vocabulary.

Conclusion

This study set out to observe the dynamic interplay between learner, text, and lexical variables at early stages of the complex process of contextual word learning, under naturalistic conditions of meaning-focused reading. Mixed-effects statistical models were fit to the data representing explicit and tacit lexical knowledge in a manner that reflected the exploratory nature of the study. Our results revealed a complex landscape of contextual L2 word learning, in which individual differences (age, L2 lexical proficiency, L1, gender, learning strategies, levels of enjoyment) and lexical and text characteristics (concreteness, frequency, distribution, and saliency of use) individually and together affect L2 lexical development from reading.

In line with general theories of learning, our study has shown that number of encounters with a new word in reading is a major predictor of gain in explicit word knowledge. However, even at multiple encounters, contextual learning is more problematic for less proficient L2 readers. Thus, while for advanced learners extensive reading may be sufficient for sustained vocabulary development, at lower proficiencies, reading needs to be supplemented with deliberate word learning and vocabulary learning strategy training. In instructed vocabulary acquisition, opportunities should be provided for developing fluency of L2 lexical processing, as our study shows that it is critical for contextual word learning (as demonstrated by the role of the WordCV predictor).

In addition, our results suggest that robust lexical representations are more likely to be established for key content words (i.e., words that are more frequent in the text than generally in the language). This means that, by chunking their readings by topic, L2 readers may facilitate the development of high-quality L2 word knowledge. Tertiary students, for example, are likely to gain good knowledge of specific-purpose vocabulary in their field of study by simply doing their course readings.

Final revised version accepted 29 November 2013

Notes

1 Following reviewer advice, an additional analysis was conducted for all three experimental conditions, comparing RTs in the control condition and the nonword-priming condition with those in the pseudoword-priming condition (Table S7 in the Supporting Information online). Predictors specific to the reading task were excluded from this analysis in order to avoid chance effects and interactions with the nonword-priming condition. This analysis showed that RTs in the pseudoword-priming condition were faster than in the control condition ($t = 1.99$ $SE = .01$) but slower than in the nonword-priming condition ($t = -1.84$ $SE = .14$). A marginally reliable difference between the two priming conditions indicates a tendency toward the establishment of formal-lexical representations for the critical pseudowords. The difference in response latencies between the pseudoword-priming condition and the two other conditions interacted with participant age of L2 acquisition: inhibitory pseudoword form-priming was predicted for participants who started learning English earlier in life; and these participants were also more likely to respond more slowly to targets preceded by pseudoword primes compared to targets preceded by nonword primes.

2 Thirty-three out of the 48 participants who took part in the reading study described in this article also participated in a replication of Experiment 1 of the Forster and Veres (1998) study, with real L2 word form-primes. In this non-masked experiment, primes were presented for 500 ms (same duration as targets). A clear PLE was observed with reliable facilitation in the nonword-priming condition ($t = 6.09$ $SE = 2.16\text{e-}04$) and reliable inhibition in the word-priming condition ($t = -2.47$ $SE = 2.22\text{e-}04$). This shows that, for the same participants, known L2 word primes inhibited the processing of their orthographic neighbour targets (at least with visible primes).

3 Although there are no direct interpretations of LSA similarity scores, the LSA website gives the following as example input: cat/mouse .34; house/dog .02.

4 An additional analysis was conducted with all three experimental conditions, comparing RTs in the pseudoword-priming condition with those in the control and the word-priming condition (Table S8, Figure S1 in the Supporting Information online). Predictors specific to the reading task were excluded from this analysis. The analysis showed that both comparisons interacted reliably with participant RTs to primes ($t = 7.02$ $SE = .01$; $t = 4.39$ $SE = .01$): slower responses to primes were associated with faster responses to targets in the pseudoword-priming condition compared to those in the word-priming and in the control condition (i.e., positive semantic priming). Faster responses to primes were likely to result in inhibitory semantic priming with pseudoword primes and facilitatory priming with word primes. This suggests a slower trajectory of semantic activation for the newly-learned pseudowords compared to known L2 words. In addition, prime response accuracy affected the difference between RTs to targets in the pseudoword- and word-priming conditions ($t = -1.82$ $SE = .03$): for correct responses to primes, responses to targets preceded by word primes were faster compared to responses to targets preceded by pseudoword primes but, for incorrect responses to primes, RTs to targets were slower in the word-priming condition than in the pseudoword-priming condition. Pseudoword priming (the difference between RTs to targets in the pseudoword and the control condition), on the other hand, was not reliably affected by the accuracy of responses to primes. The interactions of RTs to targets with response accuracy and latency of responses to primes in the word-priming condition show that within-L2 semantic priming depends on robust knowledge of meaning of the experimental stimuli. Thus, on trials where participants did not have good knowledge of the primes (made an error in identifying them as English words and were slower in making lexical decisions), RTs in the word-priming condition were not only slower than those in the pseudoword priming condition, but also slower than those in the control condition, that is, inhibition rather than facilitation was observed (cf. Dagenbach et al., 1990). The difference in RTs to targets in the pseudoword- and word-priming conditions also interacted with participant WordCV ($t = -2.21$ $SE = .01$): RTs to targets primed by pseudowords were likely to be faster than RTs to targets primed by words

for more lexically proficient participants, suggesting a learning advantage for these participants. Finally, interactions with the age of participant L2 acquisition were not statistically reliable, but the effect was in the same direction as in the separate pseudoword-priming analysis.

References

Anderson, J. R. (1982). Acquisition of cognitive skill. *Psychological Review*, *89*, 369–406.

Anderson, J. R. (1989). Human memory: An adaptive perspective. *Psychological Review*, *96*, 703–719.

Andrews, S. (2008). Lexical expertise and reading skill. In B. H. Ross (Ed.), *The psychology of learning and motivation: Advances in research and theory* (Vol. *49*, pp. 247–281). San Diego, CA: Elsevier.

Andrews, S., & Bond, R. (2009). Lexical expertise and reading skill: Bottom-up and top-down processing of lexical ambiguity. *Reading and Writing*, *22*, 687–711.

Andrews, S., & Hersch, J. (2010). Lexical precision in skilled readers: Individual differences in masked neighbor priming. *Journal of Experimental Psychology: General*, *139*, 299–318.

Andrews, S., & Lo, S. (2012). Not all skilled readers have cracked the code: Individual differences in masked form priming. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *39*, 152–163.

Baayen, R. H. (2008). *Analyzing linguistic data: A practical introduction to statistics using R.* Cambridge, UK: Cambridge University Press.

Baayen, R. H., Davidson, D. J., & Bates, D. (2008). Mixed-effects modeling with crossed random effects for subjects and items. *Journal of Memory and Language*, *59*, 390–412.

Baayen, R. H., & Milin, P. (2010). Analyzing reaction times. *International Journal of Psychological Research*, *3*, 12–28.

Baayen, R. H., Piepenbrock, R., & Gulikers, L. (1995). *The CELEX lexical database* [CD-ROM]. Philadelphia: Linguistic Data Consortium.

Balota, D. A., Yap, M. J., Cortese, M. J., Hutchison, K. A., Kessler, B., Loftis, B., et al. (2007). The English Lexicon Project. *Behavior Research Methods*, *39*, 445–459.

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*, 255–278.

Bates, D. M. (2011). *Linear mixed model implementation in lme4*. Retrieved from http://cran.r-project.org/web/packages/lme4/vignettes/Implementation.pdf

Bates, D., & Sarkar, D. (2010). *lme4*: *Linear mixed-effects models using S4 classes*. Retrieved from http://CRAN.R-project.org/package=lme4, R package version 0.999375--33

Beck, I. L., McKeown, M. G., & McCaslin, E. S. (1983). Vocabulary: All contexts are not created equal. *Elementary School Journal*, *83*, 177–181.

Belsley, D. A., Kuh, E., & Welsch, R. E. (1980). *Regression diagnostics: Identifying influential data and sources of collinearity*. New York: Wiley.

Birdsong, D. (2006). Age and second language acquisition and processing: A selective overview. *Language Learning*, *56*(Supp. 1), 9–49.

Bolger, D. J., Balass, M., Landen, E., & Perfetti, C. A. (2008). Contextual variation and definitions in learning the meanings of words. *Discourse Processes*, *45*, 122–159.

Bondi, M. (2010). Perspectives on keywords and keyness. In M. Bondi & M. Scott (Eds.), *Keyness in Texts* (pp. 1–18). Amsterdam: John Benjamins.

Borovsky, A., Kutas, M., & Elman, J. L. (2013). Getting it right: Word learning across the hemispheres. *Neuropsychologia*, *51*, 825–837.

Brown, J. I., Nelson, M. J., & Denny, E. C. (1973). *The Nelson–Denny reading test: Forms C and D for high schools and colleges*. Boston: Houghton Mifflin.

Brown, R., Waring, R., & Donkaewbua, S. (2008). Incidental vocabulary acquisition from reading, reading-while-listening, and listening. *Reading in a Foreign Language*, *20*, 136–163.

Castles, A., Davis, C., Cavalot, P., & Forster, K. (2007). Tracking the acquisition of orthographic skills in developing readers: Masked priming effects. *Journal of Experimental Child Psychology*, *97*, 165–182.

Chen, C., Xue, G., Dong, Q., Jin, Z., Li, T., Xue, F., et al. (2007). Sex determines the neurofunctional predictors of visual word learning. *Neuropsychologia*, *45*, 741–747.

Coltheart, M., Davelaar, E., Jonasson, J., & Besner, D. (1977). Access to the internal lexicon. In S. Dornic (Ed.), *Attention and performance VI* (pp. 535–555). Hillsdale, NJ: Erlbaum.

Crossley, S. A., Salsbury, T., & McNamara, D. S. (2009). Measuring L2 lexical growth using hypernymic relationships. *Language Learning*, *59*, 307–334.

Crossley, S. A., Salsbury, T., McNamara, D. S., & Jarvis, S. (2011). Predicting lexical proficiency in language learners using computational indices. *Language Testing*, *28*, 561–580.

Dagenbach, D., Carr, T. H., & Barnhardt, T. M. (1990). Inhibitory semantic priming of lexical decisions due to failure to retrieve weakly activated codes. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *16*, 328–340.

Davis, C. J., & Lupker, S. J. (2006). Masked inhibitory priming in English: Evidence for lexical inhibition. *Journal of Experimental Psychology: Human Perception and Performance*, *32*, 668–687.

Day, R. R., & Park, J. (2005). Developing reading comprehension questions. *Reading in a Foreign Language*, *17*, 60–73.

Dijkstra, T., Miwa, K., Brummelhuis, B., Sappelli, M., & Baayen, R. H. (2010). How cross-language similarity and task demands affect cognate recognition. *Journal of Memory and Language*, *62*, 284–301.

Dong, Q., Mei, L., Xue, G., Chen, C., Li, T., Xue, F., et al. (2008). Sex-dependent neurofunctional predictors of long-term maintenance of visual word learning. *Neuroscience Letters*, *430*, 87–91.
Elgort, I. (2011). Deliberate learning and vocabulary acquisition in a second language. *Language Learning*, *61*, 367–413.
Ellis, N. C. (2002). Frequency effects in language processing: A review with implications for theories of implicit and explicit language acquisition. *Studies in Second Language Acquisition*, *24*, 143–188.
Ellis, N. C., & Beaton, A. (1993). Psycholinguistic determinants of foreign language vocabulary learning. *Language Learning*, *43*, 559–617.
Ferrel Tekmen, E. A., & Daloğlu, A. (2006). An investigation of incidental vocabulary acquisition in relation to learner proficiency level and word frequency. *Foreign Language Annals*, *39*, 220–243.
Forster, K. I., & Vereš, C. (1998). The prime lexicality effect: Form priming as a function of prime awareness, lexical status, and discrimination difficulty. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *24*, 498–514.
Fraser, C. A. (1999). Lexical processing strategy use and vocabulary learning through reading. *Studies in Second Language Acquisition*, *21*, 225–241.
Frishkoff, G. A., Collins-Thompson, K., Perfetti, C. A., & Callan, J. (2008). Measuring incremental changes in word knowledge: Experimental validation and implications for learning and assessment. *Behavior Research Methods*, *40*, 907–925.
Frishkoff, G. A., Perfetti, C. A., & Collins-Thompson, K. (2010). Lexical quality in the brain: ERP evidence for robust word learning from context. *Developmental Neuropsychology*, *35*, 376–403.
Fukkink, R. G., & de Glopper, K. (1998). Effects of instruction in deriving word meanings from context: A meta-analysis. *Review of Educational Research*, *68*, 450–469.
Grabe, W. (2009). *Reading in a second language: Moving from theory to practice*. New York: Cambridge University Press.
Graesser, A. C., McNamara, D. S., Louwerse, M. M., & Cai, Z. (2004). Coh-Metrix: Analysis of text on cohesion and language. *Behavioral Research Methods, Instruments, and Computers*, *36*, 193–202.
Harrell, F. E. (2001). *Regression modeling strategies: With applications to linear models, logistic regression, and survival analysis*. New York: Springer-Verlag.
Hintzman, D. L. (1986). "Schema abstraction" in a multiple-trace memory model. *Psychological Review*, *93*, 411–428.
Horst, M., Cobb, T., & Meara, P. (1998). Beyond a clockwork orange: Acquiring second language vocabulary through reading. *Reading in a Foreign Language*, *11*, 207–223.
Hu, M., & Nation, I. S. P. (2000). Unknown vocabulary density and reading comprehension. *Reading in a Foreign Language*, *13*, 403–430.

Hulstijn, J. H. (2013). Incidental learning in second language acquisition. In C. A. Chapelle (Ed.), *The encyclopedia of applied linguistics* (pp. 2632–2637). New York: Wiley-Blackwell.

Hulstijn, J. H. (2001). Intentional and incidental second language vocabulary learning: A reappraisal of elaboration, rehearsal and automaticity. In P. Robinson (Ed.), *Cognition and second language instruction* (pp. 258–286). Cambridge, UK: Cambridge University Press.

Hulstijn, J. H., & Laufer, B. (2001). Some empirical evidence for the Involvement Load Hypothesis in vocabulary acquisition. *Language Learning*, *51*, 539–558.

Jaeger, T. F. (2010). Redundancy and reduction: Speakers manage syntactic information density. *Cognitive Psychology*, *61*, 23–62.

Jaeger, T. F. (2008). Categorical data analysis: Away from ANOVAs (transformation or not) and towards logit mixed models. *Journal of Memory and Language*, *59*, 434–446.

Jaeger, T. F., & Kuperman, V. (2009). Issues and solutions in fitting, evaluating, and interpreting regression models. Presentation slides, WOMM'09 at the CUNY Sentence Processing Conference. Retrieved from www.hlp.rochester.edu/resources/WOMM/JaegerKuperman.pdf

Joordens, S., & Becker, S. (1997). The long and short of semantic priming effects in lexical decision. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *23*, 1083–1105.

Kaushanskayaa, M., Marianb, V., & Yoo, J. (2011). Gender differences in adult word learning. *Acta Psychologica*, *137*, 24–35.

Kim, K. H. S., Relkin, N. R., Lee, K., & Hirsch, J. (1997). Distinct cortical areas associated with native and second languages. *Nature*, *388*, 171–174.

Kimura, D. (1999). *Sex and cognition*. Cambridge, MA: MIT Press.

Landauer, T. K., Foltz, P. W., & Laham, D. (1998). Introduction to latent semantic analysis. *Discourse Processes*, *25*, 259–284.

Laufer, B. (1989). What percentage of text lexis is essential for comprehension? In C. Lauren & M. Nordman (Eds.), *Special language: From humans thinking to thinking machines* (pp. 316–323). Clevedon, UK: Multilingual Matters.

Laufer, B., & Shmueli, K. (1997). Memorizing new words: Does teaching have anything to do with it? *RELC Journal*, *28*, 89–108.

Lawson, M. J., & Hogben, D. (1996). The vocabulary-learning strategies of foreign-language students. *Language Learning*, *46*, 101–135.

Levitt, S. D., & Dubner, S. J. (2006). *Freakonomics*. Camberwell, Australia: Penguin.

Masson, M. E. J. (1995). A distributed memory model of semantic priming. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *21*, 3–23.

McKeown, M. G. (1985). The acquisition of word meaning from context by children of high and low ability. *Reading Research Quarterly*, *20*, 482–496.

McRae, K., & Boisvert, S. (1998). Automatic semantic similarity priming. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *24*, 558–572.

Mestres-Misse, A., Rodriguez-Fornells, A., & Münte, T. F. (2007). Watching the brain during meaning acquisition. *Cerebral Cortex*, *17*, 1858–1866.

Meyer, D. E., & Schvaneveldt, R. W. (1971). Facilitation in recognizing pairs of words: Evidence of a dependence between retrieval operations. *Journal of Experimental Psychology*, *90*, 227–234.

Mondria, J.-A., & Wit-de Boer, M. (1991). The effects of contextual richness on the guessability and the retention of words in a foreign language. *Applied Linguistics*, *12*, 249–267.

Nassaji, H. (2003). L2 vocabulary learning from context: Strategies, knowledge sources, and their relationship with success in L2 Lexical Inferencing. *TESOL Quarterly*, *37*, 645–670.

Nation, I. S. P. (2001). *Learning vocabulary in another language*. Cambridge, UK: Cambridge University Press.

Nation, I. S. P. (2006). How large a vocabulary is needed for reading and listening? *Canadian Modern Language Review*, *63*, 59–82.

Nation, I. S. P., & Beglar, D. (2007). A vocabulary size test. *The Language Teacher*, *31*, 9–13.

Neely, J. H. (1991). Semantic priming effects in visual word recognition: A selective review of current findings and theories. In D. Besner & G. W. Humphreys (Eds.), *Basic processes in reading* (pp. 264–336). Hillsdale, NJ: Erlbaum.

Ozuru, Y., Briner, S., Kurby, C. A., & McNamara, D. S. (2013). Comparing comprehension measured by multiple-choice and open-ended questions. *Canadian Journal of Experimental Psychology*, *67*, 215–227.

Paribakht, S., & Wesche, M. (1999). Reading and “incidental” L2 vocabulary acquisition: An introspective study of lexical inferencing. *Studies in Second Language Acquisition*, *21*, 195–224.

Pellicer-Sanchez, A., & Schmitt, N. (2010). Incidental vocabulary acquisition from reading an authentic novel: Do *Things Fall Apart*? *Reading in a Foreign Language*, *22*, 31–55.

Perani, D., Paulesu, E., Galles, N. S., Dupoux, E., Dehaene, S., Bettinardi, V., et al. (1998). The bilingual brain: Proficiency and age of acquisition of the second language. *Brain*, *121*, 1841–1852.

Perfetti, C. A. (2007). Reading ability: Lexical quality to comprehension. *Scientific Studies of Reading*, *11*, 357–383.

Perfetti, C. A., Cao, F., & Booth, J. R. (2013). Specialization and universals in the development of reading skills: How Chinese research informs a universal science of reading. *Scientific Studies in Reading*, *17*, 5–21.

Perfetti, C. A., & Hart, L. (2001). The lexical bases of comprehension skill. In D. Gorfien (Ed.), *On the consequences of meaning selection* (pp. 67–86). Washington, DC: American Psychological Association.

Perfetti, C. A., & Hart, L. (2002). The lexical quality hypothesis. In L. Verhoeven, C. Elbro, & P. Reitsma (Eds.), *Precursors of functional literacy* (Vol. *11*, pp. 189–213). Amsterdam: John Benjamins.

Phillips, N. A., Segalowitz, N., O'Brien, I., & Yamasakia, N. (2004). Semantic priming in a first and second language: Evidence from reaction time variability and event-related brain potentials. *Journal of Neurolinguistics*, *17*, 237–262.

Pigada, M., & Schmitt, N. (2006). Vocabulary acquisition from extensive reading: A case study. *Reading in a Foreign Language*, *18*, 1–28.

Pulido, D. (2007). The effects of topic familiarity and passage sight vocabulary on L2 lexical inferencing and retention through reading. *Applied Linguistics*, *28*, 66–86.

Pulido, D. (2009). How involved are American L2 learners of Spanish in lexical input processing tasks during reading? *Studies in Second Language Acquisition*, *31*, 31–58.

Pulido, D., & Hambrick, D. Z. (2008). The virtuous circle: Modeling individual differences in L2 reading and vocabulary development. *Reading in a Foreign Language*, *20*, 164–190.

Rott, S. (1999). The effect of exposure frequency on intermediate language learners' incidental vocabulary acquisition and retention through reading. *Studies in Second Language Acquisition*, *21*, 589–619.

Salsbury, T., Crossley, S. A., & McNamara, D. S. (2011). Psycholinguistic word information in second language oral discourse. *Second Language Research*, *27*, 343–360.

Saragi, T., Nation, I. S. P., & Meister, G. F. (1978). Vocabulary learning and reading. *System*, *6*, 72–78.

Schmidt, R. (2001). Attention. In P. Robinson (Ed.), *Cognition and second language instruction* (pp. 3–32). Cambridge, UK: Cambridge University Press.

Segalowitz, N. (2000). Automaticity and attentional skill in fluent performance. In H. Riggenbach (Ed.), *Perspectives on fluency* (pp. 200–219). Ann Arbor: University of Michigan Press.

Segalowitz, N. S., & Segalowitz, S. J. (1993). Skilled performance, practice, and the differentiation of speed-up from automatisation effects: Evidence from second language word recognition. *Applied Psycholinguistics*, *14*, 369–385.

Silverberg, S., & Samuel, A. G. (2004). The effect of age of second language acquisition on the representation and processing of second language words. *Journal of Memory and Language*, *51*, 381–398.

Solomon, D. J. (2004). The rating reliability calculator. *BMC Medical Research Methodology*, *4*, 11.

Ullman, M. T., Estabrooke, I. V., Steinhauer, K., Brovetto, C., Pancheva, R., Ozawa, K., et al. (2002). Sex differences in the neurocognition of language. *Brain and Language*, *83*, 141–143.

Vidal, K. (2011). A comparison of the effects of reading and listening on incidental vocabulary acquisition. *Language Learning*, *61*, 219–258.

Wald, A. (1943). Test of statistical hypotheses concerning several parameters when the number of observations is large. *Transactions of the American Mathematical Society*, *54*, 426–482.
Waring, R., & Takaki, M. (2003). At what rate do learners learn and retain new vocabulary from reading a graded reader? *Reading in a Foreign Language*, *15*, 130–163.
Webb, S. (2005). Receptive and productive vocabulary learning: The effects of reading and writing on word knowledge. *Studies in Second Language Acquisition*, *27*, 33–52.
Webb, S. (2007). Learning word pairs and glossed sentences: The effects of a single context on vocabulary knowledge. *Language Teaching Research*, *11*, 63–81.
Webb, S. (2008). The effects of context on incidental vocabulary learning. *Reading in a Foreign Language*, *20*, 232–245.
Weber-Fox, C. M., & Neville, H. J. (1999). Functional neural subsystems are differentially affected by delays in second language immersion: ERP and behavioral evidence in bilinguals. In D. Birdsong (Ed.), *Second language acquisition and the critical period hypothesis* (pp. 23–38). Mahwah, NJ: Erlbaum.
Wuillemin, D., & Richardson, B. (1994). Right hemisphere involvement in processing later-learned languages in multilingualism. *Brain and Language*, *46*, 620–636.
Zahar, R., Cobb, T., & Spada, N. (2001). Acquiring vocabulary through reading: Effects of frequency and contextual richness. *Canadian Modern Language Review*, *57*, 541–572.

Appendix A: Participants' Characteristics

Participants' L1

Cantonese/Mandarin (n = 7); Filipino (n = 1); Finnish (n = 2); French Creole (n = 1); German (n = 11); Bengali/Hindi/Kannada/Tamil/Telugu/Sinhalese/Urdu (n = 5); Italian (n = 2); Japanese (n = 1); Korean (n = 3); Malay (n = 8); Polish (n = 1); Romanian (n = 1); Russian/Ukrainian (n = 3); Spanish (n = 2).

	Mean	*SD*	Range
Age	26.60	5.60	18–41
Age of L2 acquisition (L2Age)	8.40	4.50	1–26
Learning duration (yrs)	18.30	5.90	6–33
Time in an English-speaking country (yrs)	3.30	3.40	0–15

Appendix B: Materials

Table B1 Lexical profile of the text: Lexical coverage

Freq. Level	Families	Types	Tokens	Coverage%	Cumulative%
K1 Words :	892	2143	32273	79.11	79.11%
K2 Words :	674	1098	3392	8.31	87.42%
K3 Words :	383	517	1295	3.17	90.59%
K4 Words :	311	402	790	1.94	92.53%
K5 Words :	196	252	513	1.26	93.79%
K6 Words :	149	175	285	.70	94.49%
K7 Words :	111	125	238	.58	95.07%

Table B2 Pseudoword characteristics

	Mean	*SD*
Constrained bigram frequency per million	797.90	500.20
Constrained trigram frequency per million	129.80	154.80
Number of letters	5.50	.50

Pseudowords

banity, dault, creach, recibe, dello, shrick, grude, afuse, extel, tenoy, injent, failor, gence, infate, lidger, lacent, flane, pogue, piting, outlad, seacon, merial, staim, cluff, detise, frine, taive, surmit, lastor, evotic, wockey, duckle, advern, trimp, toxit, emback, pleak, medio, nurge, merly, pelon, decop, punse, puriny, modium, snall, alirn, totate.

Table B3 Comprehension questions

Chapter 1: What do schoolteachers and sumo wrestlers have in common?
Explain how the introduction of a lateness fine at a day care center may have altered the motivations of the parents.
What technique did Levitt use to uncover evidence of some teachers behaving dishonestly on standardized tests?
What evidence do the authors offer in support of the claim that some Japanese sumo wrestlers probably “throw” some of their matches?
Chapter 2: How is the Ku Klux Klan like a group of real-estate agents?
Explain Stetson Kennedy’s role in the Ku Klux Klan’s decline in popularity in the South.
What evidence do the authors offer to support their claim that an information asymmetry can be used to clients’ disadvantage in the business of selling houses?

(Continued)

Table B3 Continued

Chapter 3: Why do drug dealers still live with their moms?
Describe, in general terms, the organizational structure of the Black Disciples gang. How is it similar to the organizational structure of most businesses?
How did crack cocaine change the way urban street gangs operated?
Chapter 4: Where have all the criminals gone?
How do the authors justify why some of the explanations of the drop in the crime that occurred in the 1990s are valid and some are not? (Provide examples, if possible).

Appendix C: Descriptive Statistics: Participant and Item Predictors

		Median	Mean	*SD*	Range
Participant predictors	VST	10150	10135	1618	6900–13100
	WordCV*	.30	.30	.09	.16–.50
	Prior knowledge about economics	14	14.10	2.70	8–20
	Text comprehension	14	14.79	4.33	5–24
	Rating of interest/enjoyment (Attitude)	4.20	4.10	.60	2.6–5
	Vocabulary learning strategies	10	10.50	2.12	7–15
Item predictors	Rate of occurrence	12	18.90	19.75	6–88
	Keyness index	6.43	29.20	63.09	1–328.33
	Spacing across chapters (Range)	3.04	3.00	1.11	1–5
	Concreteness/ Abstractness (from Coh-Metrix)**	5.73	5.54	2.29	1–11

*Lower WordCV values indicate more fluent access to L2 lexical representations, that is, higher L2 lexical proficiency

**Higher values indicate words with a higher number of hypernym levels in the WordNet lexical system (i.e., words denoting more concrete concepts).

Note 1: The *Abstractness* and *Keyness* data were obtained for the original words replaced in the text by the pseudowords

Note 2: Descriptive statistics for *Gender* and *Ease of guessing from context* are reported in-text; Age, L2Age and L1 are reported in Appendix A.

Appendix D: Analysis of the Explicit Knowledge of Meaning Data: Results Summary

Fixed Effects:

	Coef.β	SE(β)	z	p	
(Intercept)	−4.24	.64	−6.59	4.6E-11	***
WordCV	−.87	.24	−3.57	3.6E-04	***
VST	.35	.27	1.30	.19	
Age	−.36	.22	−1.59	.11	
L2Age	−.08	.21	−.38	.70	
Gender	.73	.29	2.54	.01	*
CompScore	1.18	.29	4.01	6.1E-05	***
Attitude	.92	.22	4.14	3.5E-05	***
PriorKn	.29	.23	1.28	.20	
Strat	.84	.21	4.08	4.5E-05	***
RateOccur	2.90	.42	6.85	7.2E-12	***
Keyness	1.54	.39	3.95	7.8E-05	***
Range	−.91	.41	−2.24	.03	*
Concr	1.19	.44	2.72	.01	**
Letter = Final	.51	1.10	.47	.64	
Letter = Other	−1.66	.80	−2.08	.04	*
RateOccur: VST	.37	.17	2.21	.03	*
RateOccur: WordCV	.16	.12	1.34	.18	
RateOccur: CompScore	.62	.17	3.63	2.9E-04	***
WordCV: Range	−.29	.14	−2.05	.04	*

***$p < .001$; **$p < .01$; *$p < .05$.

Random effects:

Groups/ Name	Variance	*SD*	Corr							
Participant										
(Int)	2.43	1.56								
Concr	.43	.66	−.18							
Ltr = F	2.12	1.45	−.66	.49						
Ltr = O	.56	.75	−.03	−.69	.21					
Range	.39	.62	.36	.04	.40	.56				
Item										
(Int)	6.01	2.45								
VST	.48	.69	−.05							
Gndr	.38	.62	−.37	.11						
L2Age	.00	.04	.47	−.22	.61					
Age	.21	.45	.30	.45	.70	.77				
Strat	.04	.21	−.17	.90	.54	.12	.72			
Attd	.08	.29	−.09	−.30	.42	.41	.21	−.06		
PrKn	.20	.44	−.44	−.64	−.46	−.60	−.94	−.77	.03	
CmpScr	.25	.50	−.93	.39	.31	−.56	−.18	.44	.05	.23

Appendix E: Priming Experiments

Table E1 Masked form-priming experiment: Conditions and examples of stimuli

Related pseudoword prime	Related nonword prime	Unrelated prime (control)	TARGET
flane*	flave	drown	FLAKE

* *flane* means eliminate

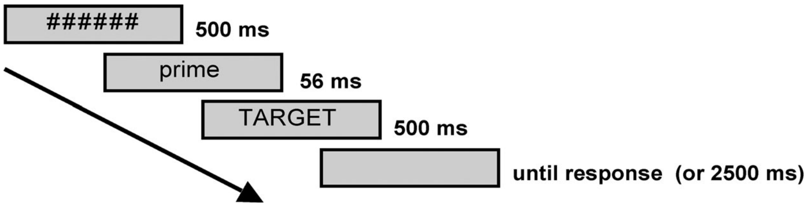

Figure E1 Masked form-priming procedure.

Table E2 Semantic priming experiment: Conditions and examples of stimuli

Related pseudoword prime	Related word prime	Unrelated prime (control)	Target	Relationship
outlad*	expert	ballot	amateur	semantic feature (-)
flane*	eliminate	frown	contestant	associative/cooccurrence

*outlad means expert; flane means eliminate

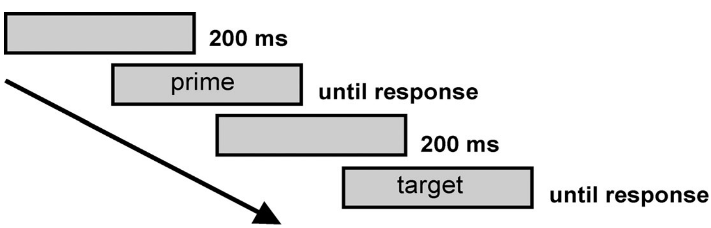

Figure E2 Semantic priming procedure.

Appendix F: Analysis of Response Times in the Masked Form-Priming Experiment

Table F1 Form-priming with pseudoword primes: Results summary
Fixed effects:

	Coef.β	SE(β)	t value	
(Intercept)	−1.23	.03	−35.79	*
Condition = pseudoword	−.02	.01	−1.84	
WordCV	.10	.03	3.17	*
VST	−.09	.03	−2.97	*
Age	.11	.03	4.01	*
L2Age	.03	.03	.88	
L1(alpha)	.09	.03	2.89	*
CompScore	−.01	.04	−.13	
Strat	.01	.03	.32	
RateOccur	−.00	.02	−.09	
Keyness	.02	.02	.86	
Range	−.03	.02	−1.36	
GuessRTG	.05	.02	2.21	*
Concr	.02	.02	1.15	
PrevRT	.03	.01	3.36	*
Trial	−.00	.02	−.18	
Condition = pseudoword:WordCV	.02	.01	1.29	
Condition = pseudoword:CompScore	−.03	.02	−1.46	
Condition = pseudoword:L2Age	−.05	.02	−3.13	*
Condition = pseudoword:L1(alpha)	.01	.01	.87	
Condition = pseudoword:Strat	.01	.01	.69	
Condition = pseudoword:Range	.01	.01	.98	
Condition = pseudoword:Concr	.02	.01	1.76	
Condition = pseudoword:GuessRTGR	−.00	.02	−.14	

Note: t > 2 indicates a significant effect at $p < .05$.

Random effects:

Groups/Name	Variance	*SD*	Corr						
Participant									
Intercpt	3.9E-02	.20							
Keyness	5.2E-04	.02	−.38						
Concr	4.7E-04	.02	−.29	−.78					
GuessRTG	3.8E-04	.02	−.97	.17	.49				
PrevRT	3.0E-05	.01	.32	.76	−.99	−.52			
Trial	2.0E-03	.05	−.12	−.87	.99	.34	−.98		
Target									
Intercpt	1.2E-02	.11							
VST	1.3E-03	.04	.31						
WordCV	1.8E-03	.04	.03	.25					
CompScre	1.7E-03	.04	−.40	−.38	−.56				
Gender	6.3E-03	.08	.29	.29	−.08	.14			
Age	6.0E-04	.03	−.13	−.70	.37	.19	−.37		
L1(alph)	6.9E-04	.03	.18	.92	.35	−.29	−.03	−.47	
Strat	3.9E-04	.02	.02	.36	.85	−.19	.34	.27	.38
Residual	5.3E-02	.23							

Table F2 Form-priming with nonword primes: Results summary
Fixed effects:

	Coef.β	SE(β)	*t* value	
(Intercept)	−1.22	.03	−37.52	*
Condition = nonword	−.05	.01	−3.79	*
VST	−.09	.03	−2.92	*
WordCV	.09	.03	3.29	*
Age	.10	.03	3.66	*
L1(alpha)	.07	.03	2.56	*
T.FreqHAL	−.03	.02	−1.77	
T.PhonN	.02	.01	1.97	
PrevRT	.04	.01	4.01	*

Note: t > 2 indicates a significant effect at *p* < .05.

Random effects:

Groups	Name	Variance	*SD*	Corr			
Participant	(Intercept)	.03	.18				
	PrevRT	1.9E-04	.01	.33			
	Trial	.00	.05	.04	−.93		
Target	(Intercept)	.01	.11				
	VST	.00	.03	.41			
	WordCV	2.8E-04	.02	.98	.24		
	Age	2.3E-04	.02	−.73	−.92	−.60	
	L1(Alpha)	.00	.03	.39	.22	.37	−.33
Residual		.06	.24				

Appendix G: Analysis of Response Times in the Semantic Priming Experiment

Table G1 Semantic priming with pseudoword primes: Results summary
Fixed effects:

	Coef.β	SE(β)	*t* value	
(Intercept)	−1.37	.03	−42.44	*
Condition = pseudoword	−.02	.02	−1.24	
VST	−.05	.03	−1.88	
WordCV	.08	.03	2.91	*
L2Age	.03	.03	1.07	
Strat	.02	.03	.89	
Compr	.01	.03	.40	
Keyness	.01	.01	.77	
Range	−.03	.01	−2.39	*
T.FreqHAL	−.05	.01	−3.91	*
T.nol	.02	.02	1.22	
PrevRT	.07	.01	7.36	*
Prime.ACC	−.11	.02	−4.67	*
Trial	−.02	.01	−1.62	
Condition = pseudoword:L2Age	.02	.01	2.40	*
Condition = pseudoword:Strat	−.02	.01	−1.92	
Condition = pseudoword:Compr	−.02	.01	−1.91	
Condition = pseudoword:Keyness	−.01	.01	−1.61	
Condition = pseudoword:PrevRT	−.04	.01	−4.18	*
Condition = pseudoword:Prime.ACC	.03	.02	1.31	

Note: $t > 2$ indicates a significant effect at $p < .05$.

Random effects:

Groups	Name	Variance	*SD*	Corr				
Participant	Intercept	.04	.19					
	T.FreqHAL	3.2E-04	.02	−.15				
	T.nol	.00	.03	.62	.06			
	PrevRT	.00	.04	.13	.36	.08		
	Prime.ACC	.02	.12	.03	−.05	.02	.37	
	Trial	3.7E-04	.02	.58	−.05	.70	.67	.17
Target	Intercept	.01	.08					
	L2Age	.00	.04	.13				
	VST	1.3E-04	.01	−.18	.95			
	WordCV	2.0E-04	.01	−.23	.59	.65		
	Strat	4.9E-04	.02	−.65	.44	.64	.07	
	Compr	1.1E-04	.01	−.19	.71	.77	.02	.86
Residual		.07	.26					

Table G2 Semantic priming with word primes: Results summary
Fixed effects:

	Coef.β	SE(β)	*t* value	
(Intercept)	−1.35	.03	−43.92	*
Condition = word	−.02	.01	−2.06	*
L2Age	.04	.03	1.74	
WordCV	.04	.03	1.68	
VST	−.04	.03	−1.61	
L1(alpha)	−.04	.03	−1.66	
T.nol	.04	.01	3.24	*
T.FreqHAL	−.04	.01	−2.97	*
PrevRT	.08	.01	9.70	*
Prime.ACC	−.13	.03	−4.92	*

Note: $t > 2$ indicates a significant effect at $p < .05$.

Random effects:

Groups	Name	Variance	*SD*	Corr			
Participant	(Intercept)	.03	.18				
	T.FreqHAL	1.6E-04	.01	−.52			
	T.nol	.00	.02	.39	−.20		
	PrevRT	.00	.05	.00	−.38	.02	
	Prime.ACC	.03	.16	−.25	−.26	.35	.22
Target	(Intercept)	.01	.08				
	L2Age	4.4E-04	.02	.27			
	VST	3.7E-04	.02	.46	.30		
	WordCV	2.1E-04	.02	−.23	.60	−.15	
	L1(alpha)	4.2E-04	.02	.59	.78	.71	.03
Residual		.06	.25				

Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Additional Data